

MAHIDOL UNIVERSITY

Faculty of Information and Communication Technology



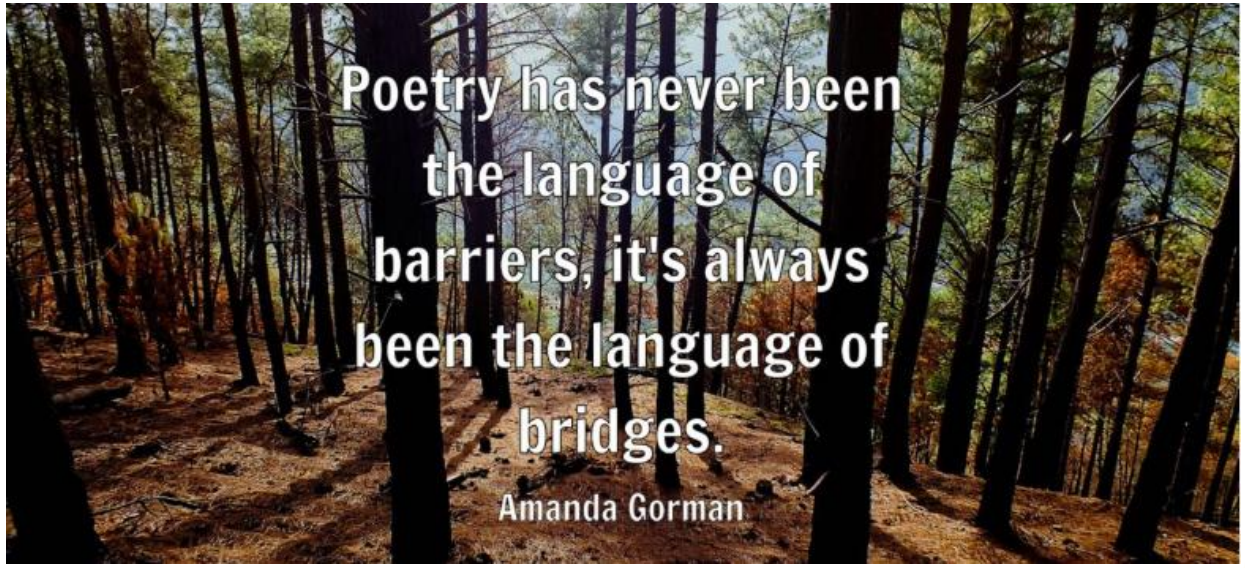
# Lecture 9
# Introduction to Data Link Layer

October 2026

ปัญญาของแผ่นดิน Wisdom of the Land

# Quote of the Week




Amanda Gorman

2021: “For there is always light, if only we're brave enough to see it. If only we're brave enough to be it.”

- Gorman reminds us that even in dark or difficult times, those positive forces always exist — they may just be hidden.
- It takes **courage** to look for hope when the world feels heavy or negative.
- This is the next level: not just seeing the light, but **becoming** it ourselves.
- It’s a call to **act** — to be the source of kindness, hope, and change that we wish to see in the world.

https://www.etsy.com/

https://www.brainyquote.com/authors/amanda-gorman-quotes
https://www.brainyquote.com/quotes/amanda_gorman_1169490?img=4

มหาวิทยาลัยมหิดล
Mahidol University

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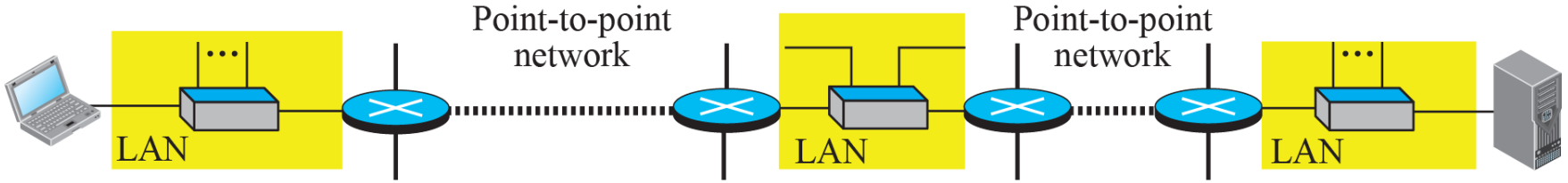
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# Data Link Layer

- In the physical layer, we focus on the physical components of networks: signals and medium (channels)
- In the data link layer, we focus on the link aspects of networks.
  - ✓ A link is a connection between two entities (nodes)
  - ✓ Each link and each node do not have to be the same type (e.g. wired and wireless).




Abstract view of an end-to-end connection




Actual view of an end-to-end connection

Note:
A traveler example with different transportation tools from ICT Salaya to Siam Paragorn via bicycle, bus 515, walk, BTS, walk

# Communication at the data-link layer

Same Network

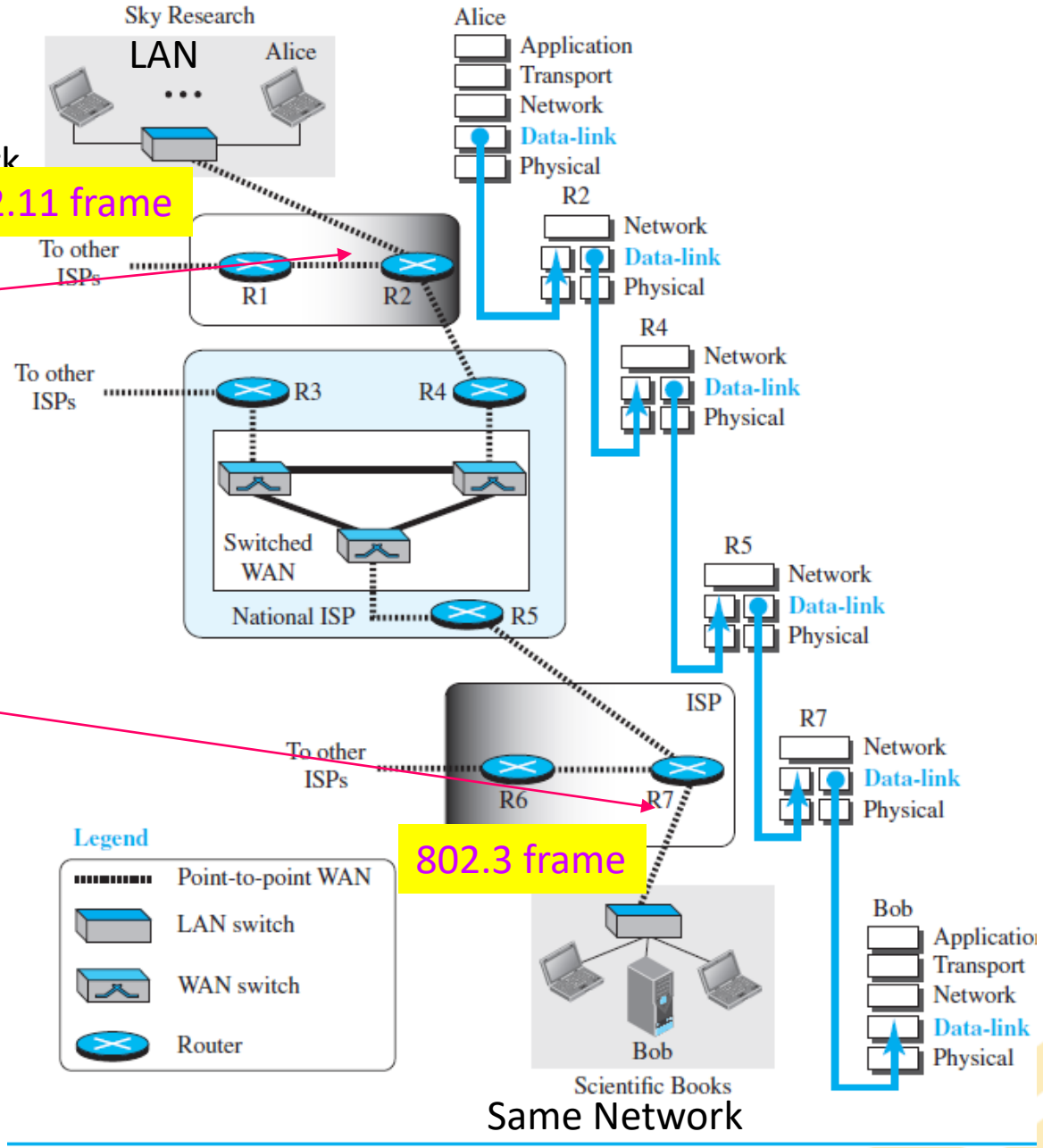
802.11 frame

Frames/Packets from hosts
need to go to IP (default) Gateway
which is at the router in the same network of hosts

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::cecd:f45b:b567:66b0%13
   IPv4 Address. . . . . . . . . . . : 192.168.0.104
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.0.1
```

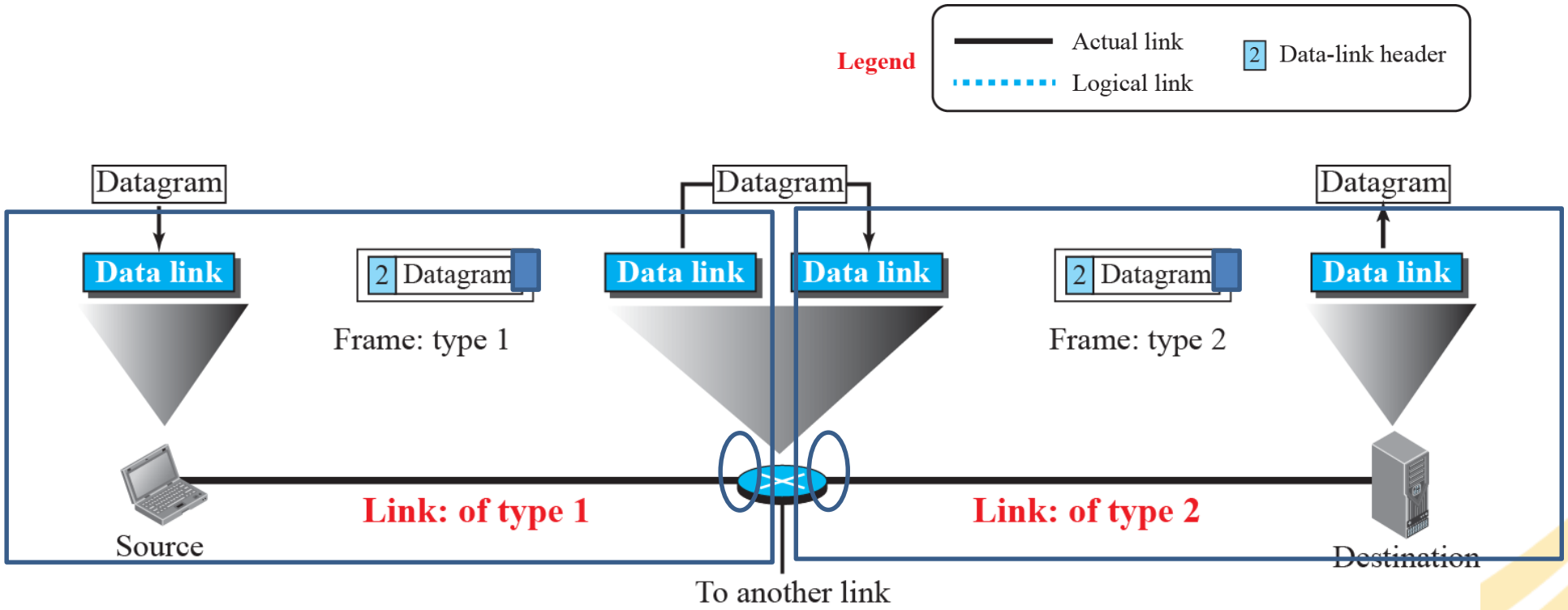





Same Network

# Data Link Layer

- The network layer passes an IP packet (datagram) to the data link layer.
- The sender encapsulates the packet (datagram) in a frame suitable for the current link.
- At the next router, that frame is removed and a new frame is created for the next link.
- The physical layer transmits the frame as signals over the medium.

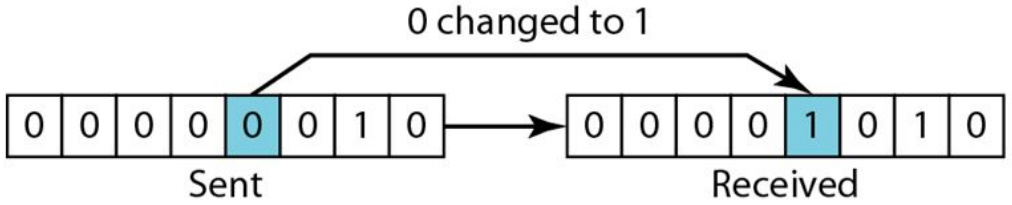


# Data Link Layer

- The physical layer takes care of signal transmission between two devices.
- Issues:
  - ✓ Error: The signals received may be in error (frame drops or out of order arrivals). The physical layer does not have any methods to fix the errors.
  - ✓ Multiple access: The sharing of medium through Frequency Division Multiplexing (FDM) and Time Division Multiplexing (TDM) is not efficient. It does not support the on-demand transmission of data from devices and the joining and leaving of devices.
  - ✓ Flow control: The physical layer cannot adaptively control the signal rate (or data rate).
- Solutions:
  - ✓ The data link layer is designed to solve these two issues (error and multiple access).

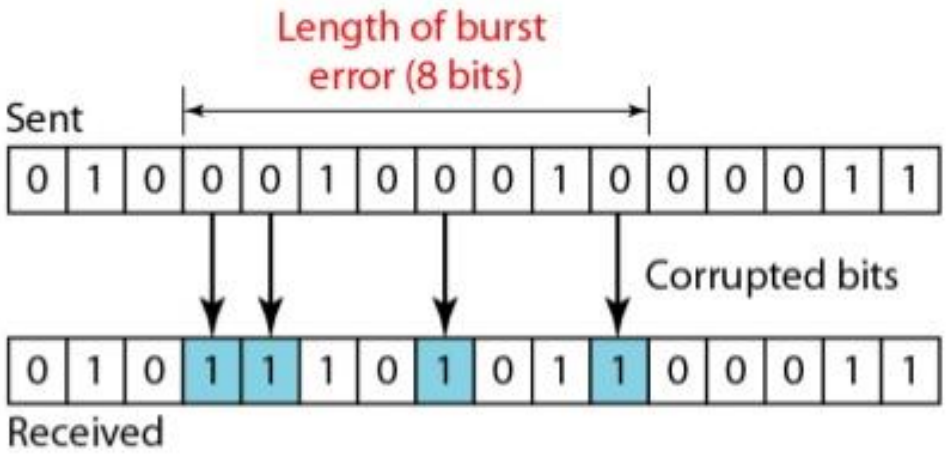
# Error

- **Single bit error**






- **Burst error**




- **Can we do anything with these types of errors?**

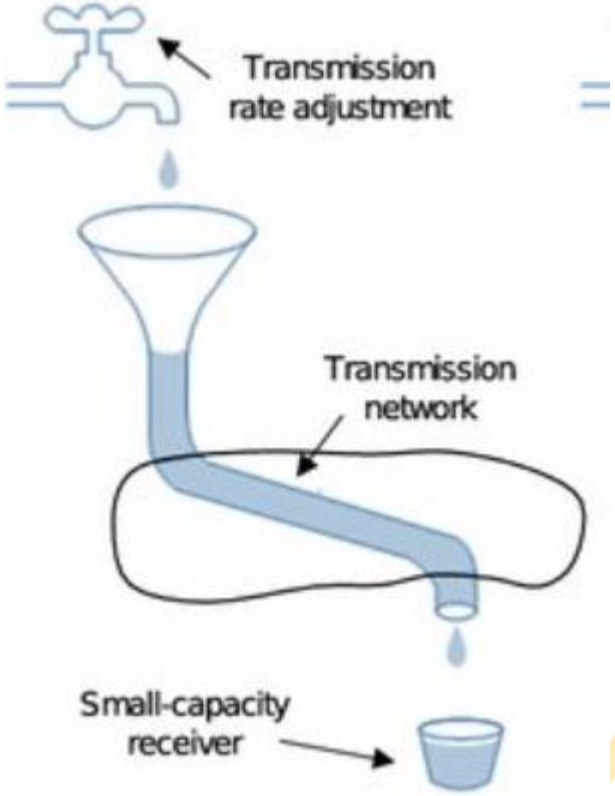
# Error Control

- When data is sent from the sender to the receiver, some data bits may be in error.
- The receiver must know how to detect such errors (error detection).
- Some receivers even know how to correct the errors received (error correction) FEC (Forward Error Correction) done by receiver
- Receiver and sender may negotiate in order to detect error and decide what to do.

# Flow Control

- Problems happen if the flow cannot be controlled



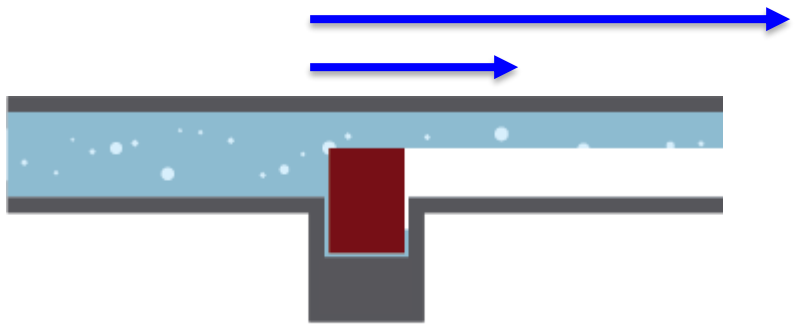



- **Can we do anything against the flooding of data?**

# Flow Control

- When data is sent from the sender to the receiver, some data bits may not be received because the receiver cannot receive any more data (e.g. out of memory, buffer overflow).
- Negotiation between the sender and receiver is required in order to make sure that the sender does not overwhelm the receiver.

Note: If overflow then new coming frame(s) need to be dropped

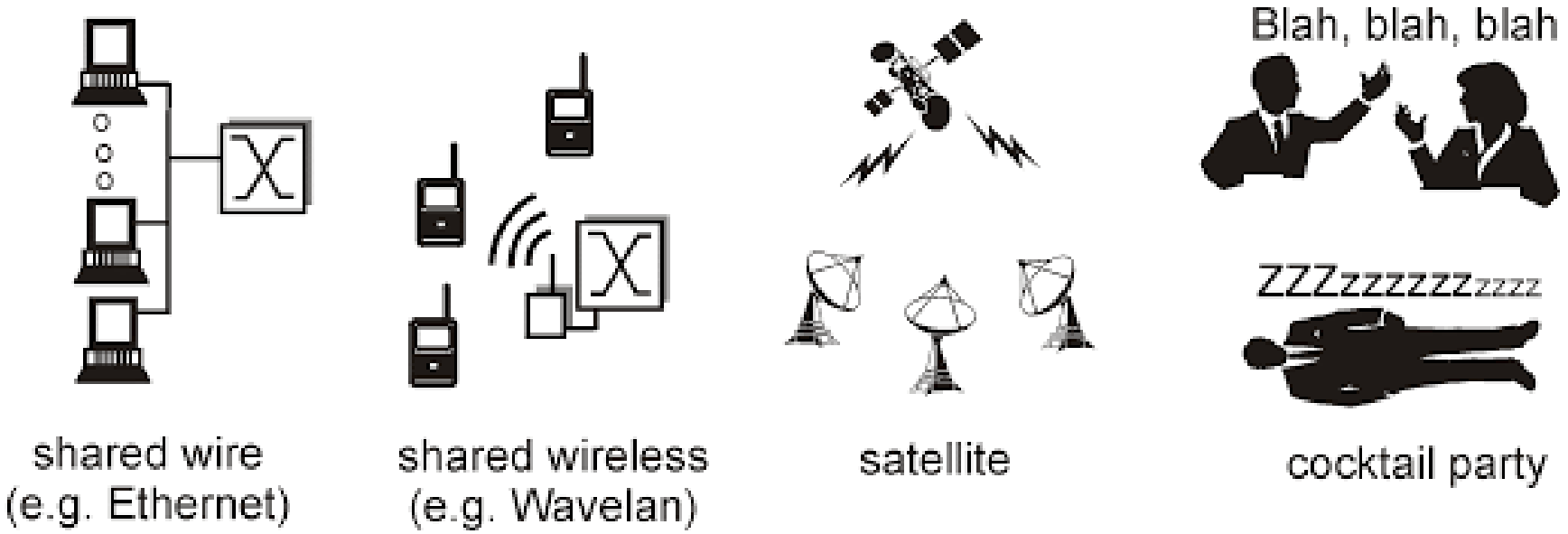


Flow Control



# Multiple Access

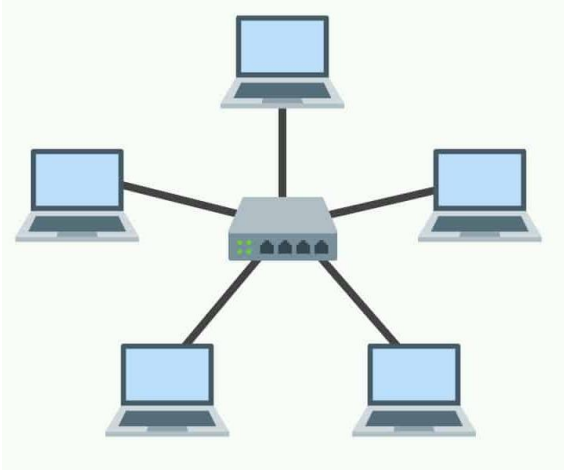
- The issue of having multiple devices trying to communicate




- **How to make sure that everyone can communicate fairly and efficiently?**

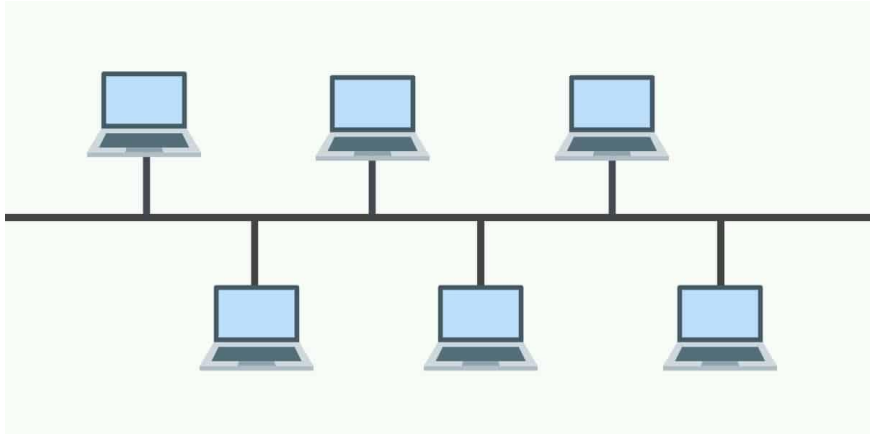
# Dedicated vs Shared Links

Full-duplex switch



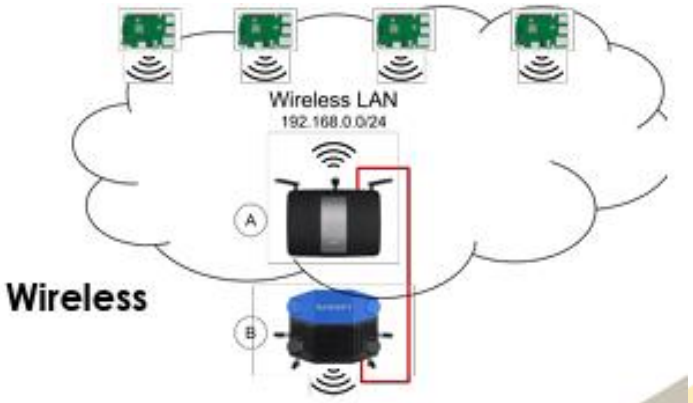
Star topology

Dedicated



Bus topology

Shared

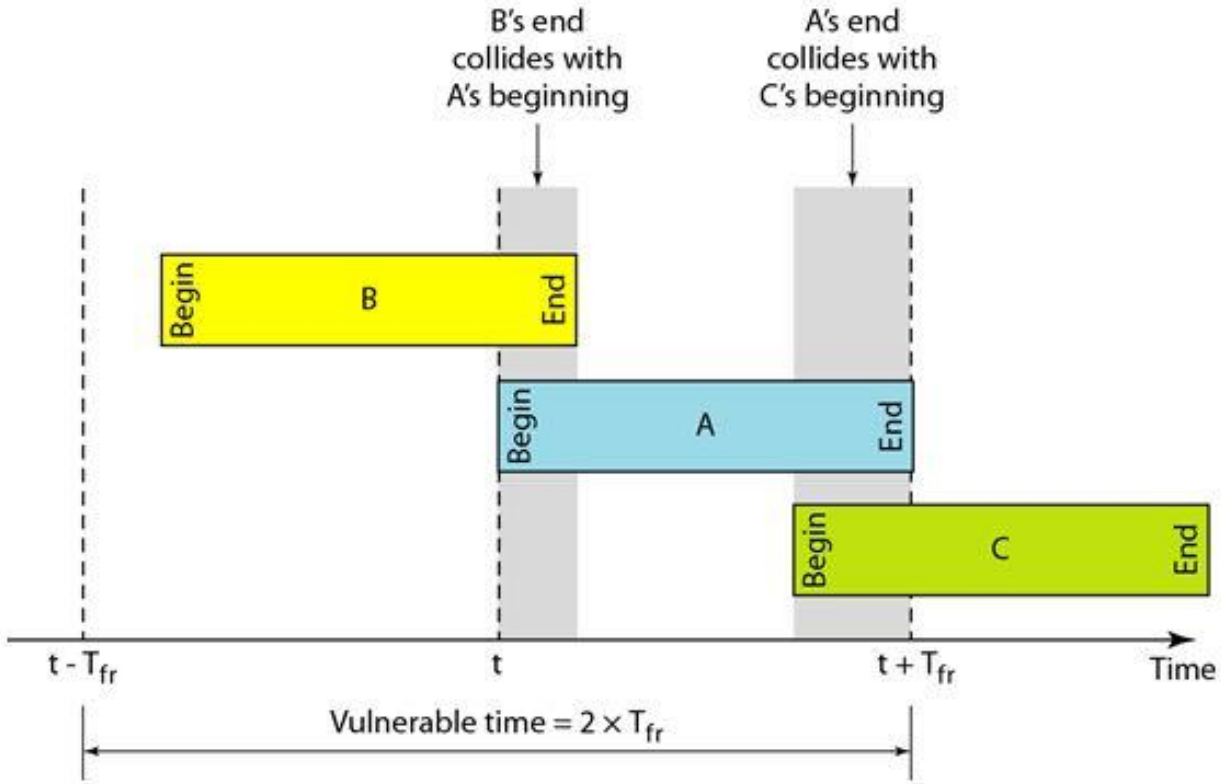



Wireless

# Media Access Control

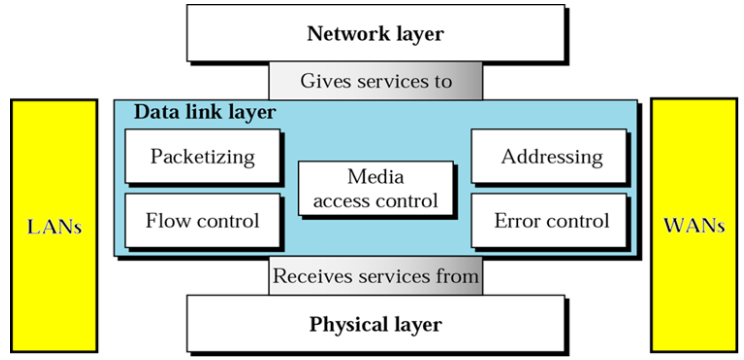
- When a medium is shared by multiple stations, without FDM or TDM, each station will compete for access to the medium to transmit data.

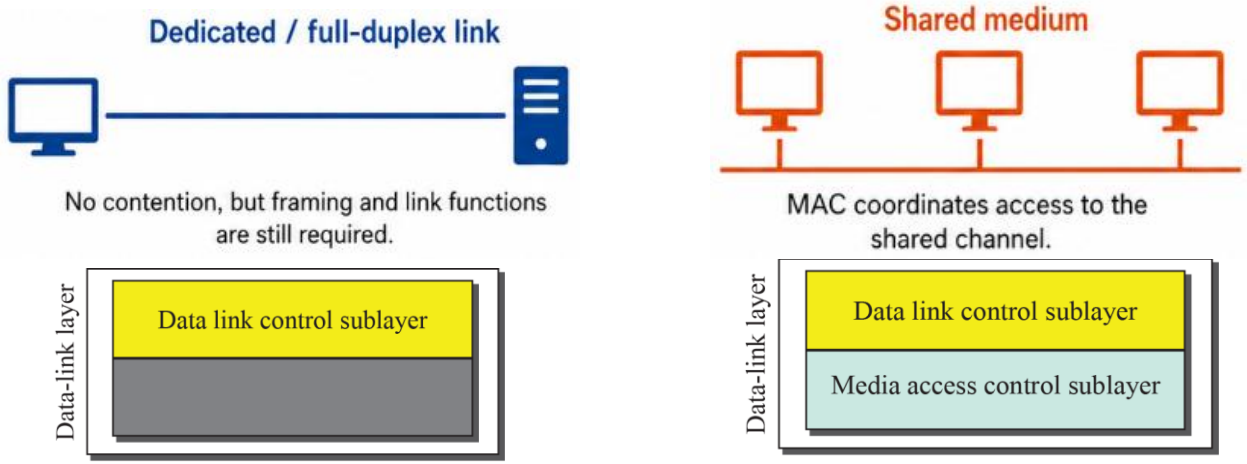
  Collision can happen.
- Media access control is a mechanism to control how a station can access the medium.

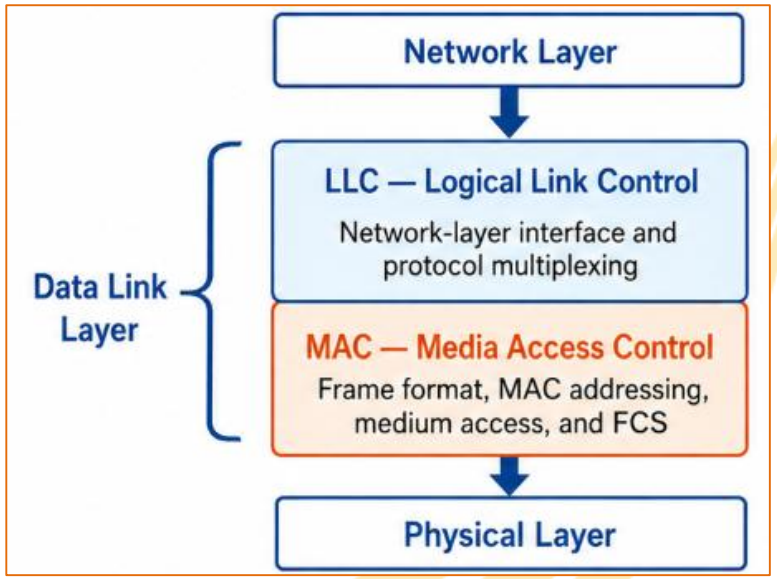


# Data Link Sublayers

- Network-layer interface [LLC]: multiplexes upper-layer protocols
- Framing: MAC in IEEE 802; performed directly by PPP/HDLC
- Link addressing [MAC]: identifies devices on the local link
- Error detection: FCS/CRC in the MAC or link protocol
- Error recovery [optional]: ACKs and retransmission
- Flow control [optional]: regulates the sender rate
- Medium access [MAC]: coordinates stations sharing a medium



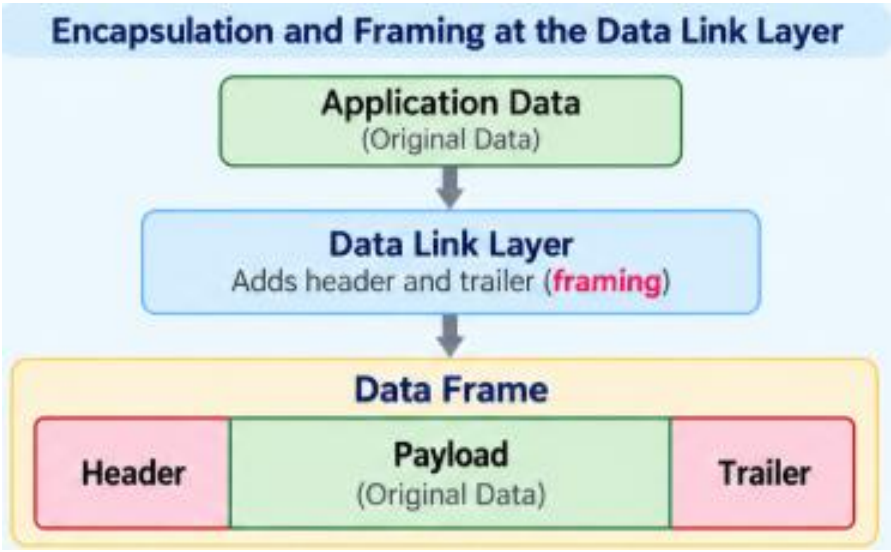



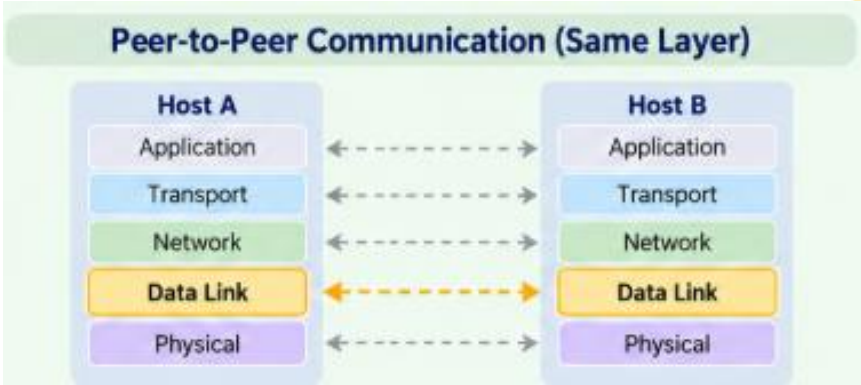




https://www.youtube.com/@JimKurose/videos

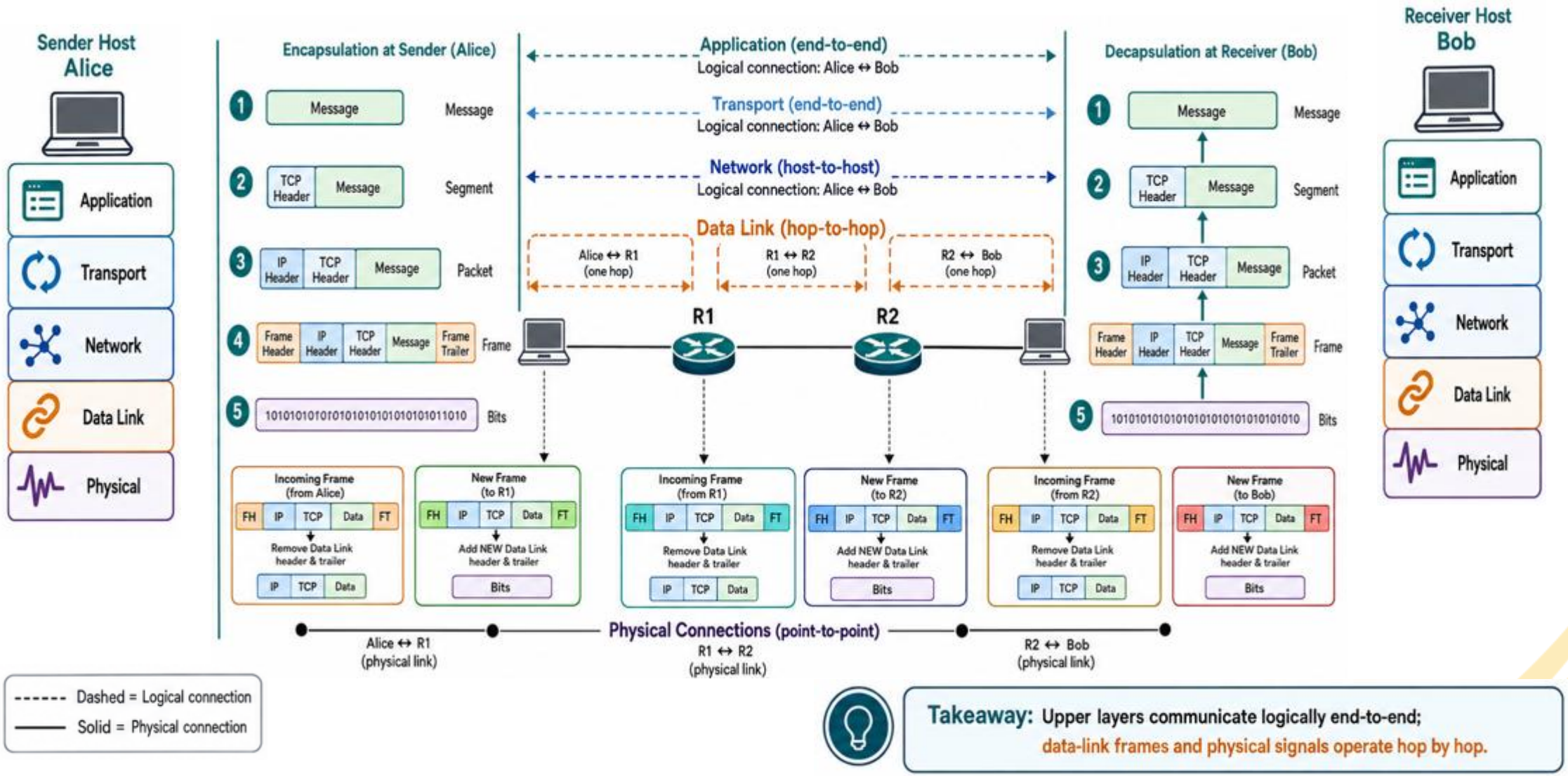
# Data Encapsulation and Framing

- In order to control the flow of data, error, and medium access information among communicating parties, extra information must be exchanged among them. These extra information must be included in data frames (**framing**). This focus is on the "**syntax** and **semantic**" parts of a protocol.
- This extra information is added to the original data as an extra block of information called '**header**' and '**tailer**'.
- Each OSI or TCP/IP layer has its own header used for peer-to-peer communicating with the communicating parties on the other side in the same layer.
- The header defines what type of control data to be exchanged. It supports communication between the two communicating parties by defining a communication language ('**protocol**').
- The original data is **encapsulated** inside a frame with a frame header.



# Recap: TCP/IP Communication Model



# Data Link Protocol (Frame) in Wireshark

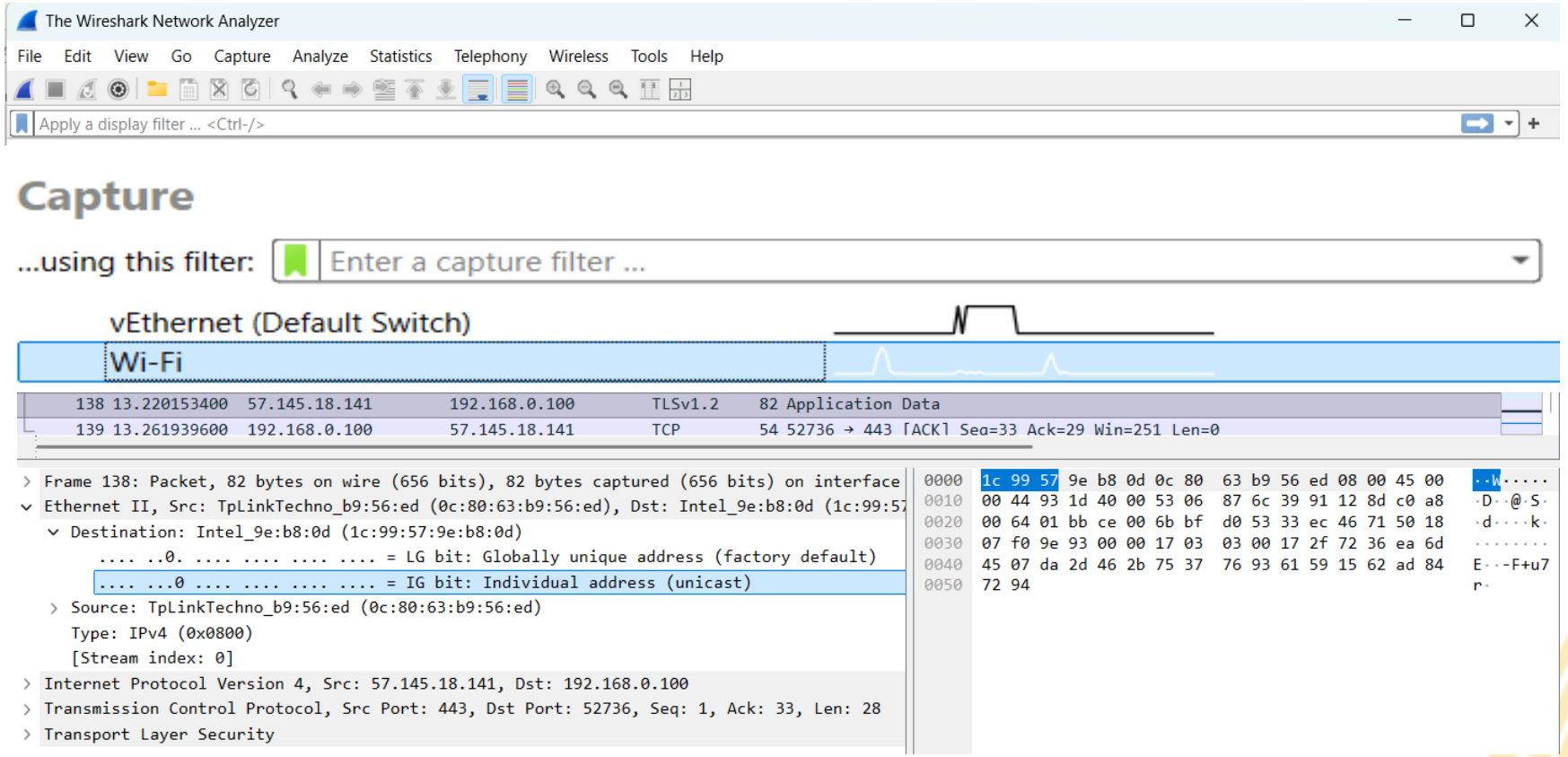



1. **Framing encapsulates** a network-layer packet for transmission over one local link. Vs decapsulates

2. The **frame header/trailer header** commonly carries **source and destination link-layer addresses.**

3. The **trailer** commonly carries a **frame check sequence (FCS)** for **error detection**.

4. At each router, the incoming frame is removed and **a new frame is created for the next link**.




Note: Wireshark normally does not show the Ethernet preamble, Start Frame Delimiter (SFD), and Frame Check Sequence (FCS)

https://www.youtube.com/watch?v=ITFG4HLpxY8

https://www.youtube.com/watch?v=xoJyT78s2UI&t=210s

# Data Link Protocols

| Link type | Example | Architecture | Key link function |
|---|---|---|---|
| Shared LAN | Ethernet 802.3 | LLC/SNAP + MAC | Framing, addressing, FCS |
| Wireless LAN | Wi-Fi 802.11 | LLC/SNAP + MAC | Access, ACK, retransmission |
| Token LAN | Token Ring 802.5 | LLC + MAC | Token access (legacy) |
| Low-power WPAN | IEEE 802.15.4 | Adaptation + MAC | Low-power framing/access |
| Point-to-point | PPP | Single data-link protocol | Framing, LCP/NCP, FCS |
| Point-to-point | HDLC | Single data-link protocol | Framing and link control |
| X.25 WAN | LAPB | Single data-link protocol | Reliable HDLC-derived link |
| Cellular radio | LTE / 5G | System-specific stack | PDCP, RLC, and MAC |

Architecture

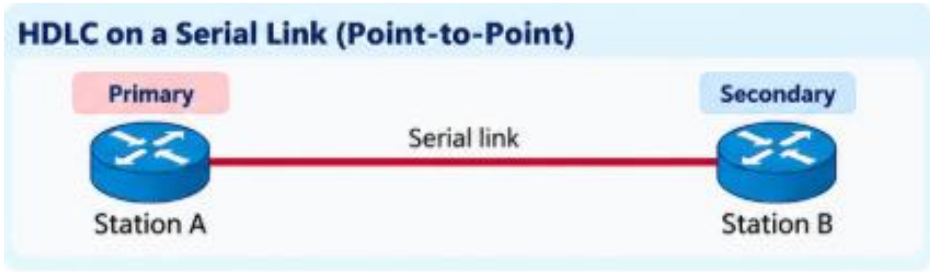
- LLC/SNAP: network-layer interface and multiplexing
- MAC: framing, addressing, FCS, and medium access
- Single data-link protocol: perform framing directly

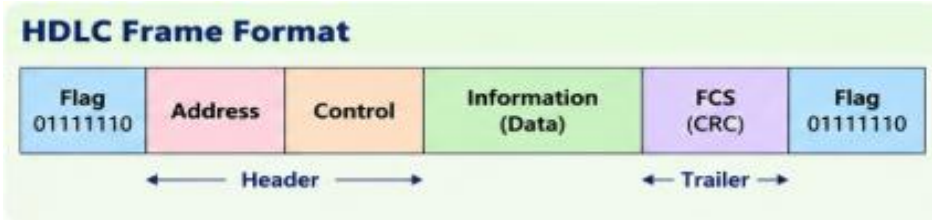
# HDLC Protocol (Serial Connection)

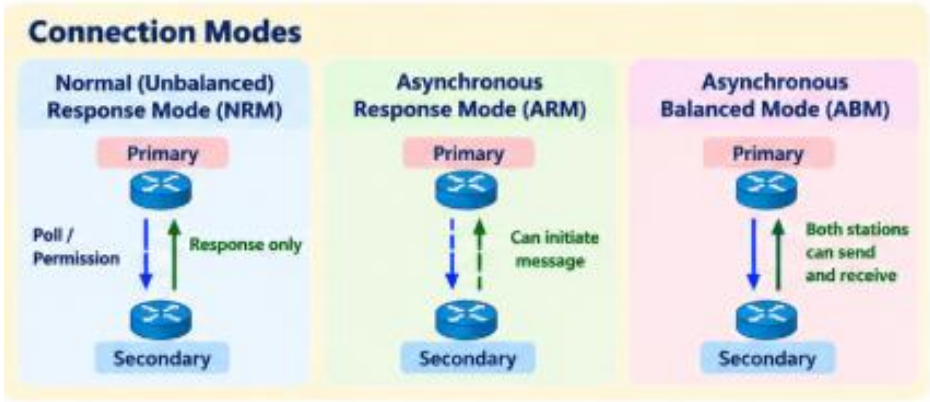
Primary: master
Secondary: slave

- HDLC (High-level Data Link Control) protocol, an **ISO*** standard, is a fundamental data link layer (layer 2) protocol used for reliable data transmission over a serial link.
- HDLC embeds control information in a data frame to support flow control and error control.
- For any HDLC communications session, one station is designated primary and the other secondary.
- A session can use one of the following connection modes, which determine how the primary and secondary stations interact.
  - ✓ **Normal (unbalanced) Respond Mode (NRM)**: The primary station **must** grant permission to the secondary stations before they can transmit data to the primary station. The secondary station responds only to the primary station. For example: SDLC.
  - ✓ **Asynchronous Response Mode (ARM)**: The secondary station can initiate a message.
  - ✓ **Asynchronous Balanced Mode (ABM):** Both stations can send and receive over a (full) duplex line.
    Nodes are hybrids, which can act either as the primary or the secondary.

| Item | Serial Connection (HDLC) | Parallel |
|---|---|---|
| Data transmission | Serial (one bit at a time) | Parallel (multiple bits at a time) |
| Typical use | Point-to-point link | Within a device or short distance |
| Protocol example | HDLC (ISO standard), PPP | e.g., parallel bus (not HDLC) |
| Main focus | Flow control, error control, reliable data transmission | High-speed local data transfer |

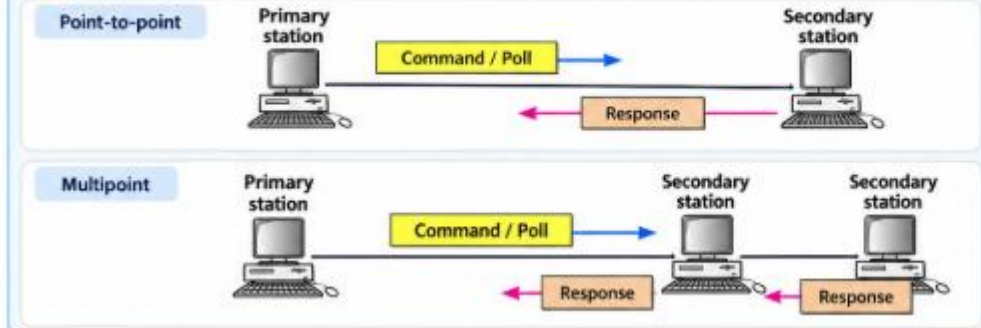


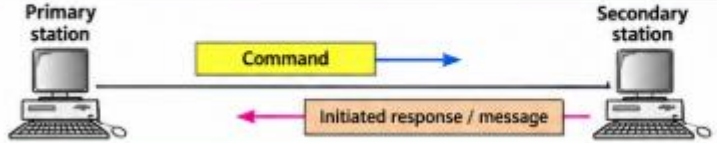
# HDLC Communication

## 1 Normal Response Mode (NRM) / Unbalanced



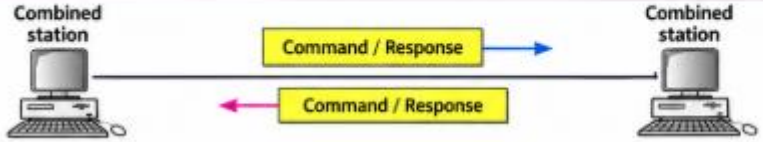

- Primary controls communication; secondary responds only after being polled or permitted.

## 2 Asynchronous Response Mode (ARM) / Unbalanced




- Secondary may initiate transmission, but the primary station still manages the link.

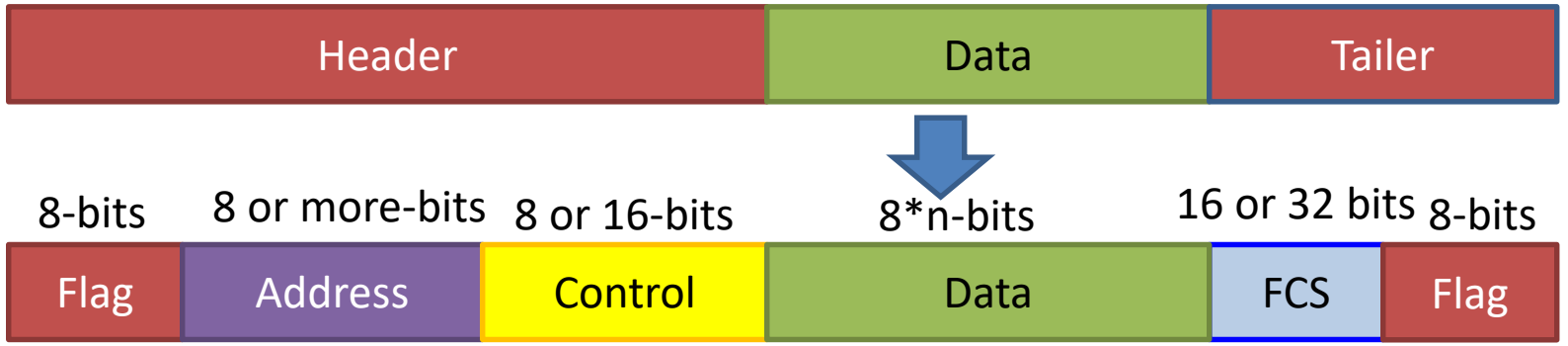
## 3 Asynchronous Balanced Mode (ABM) / Balanced




- Both stations are equal peers and can initiate transmission.

# HDLC Header

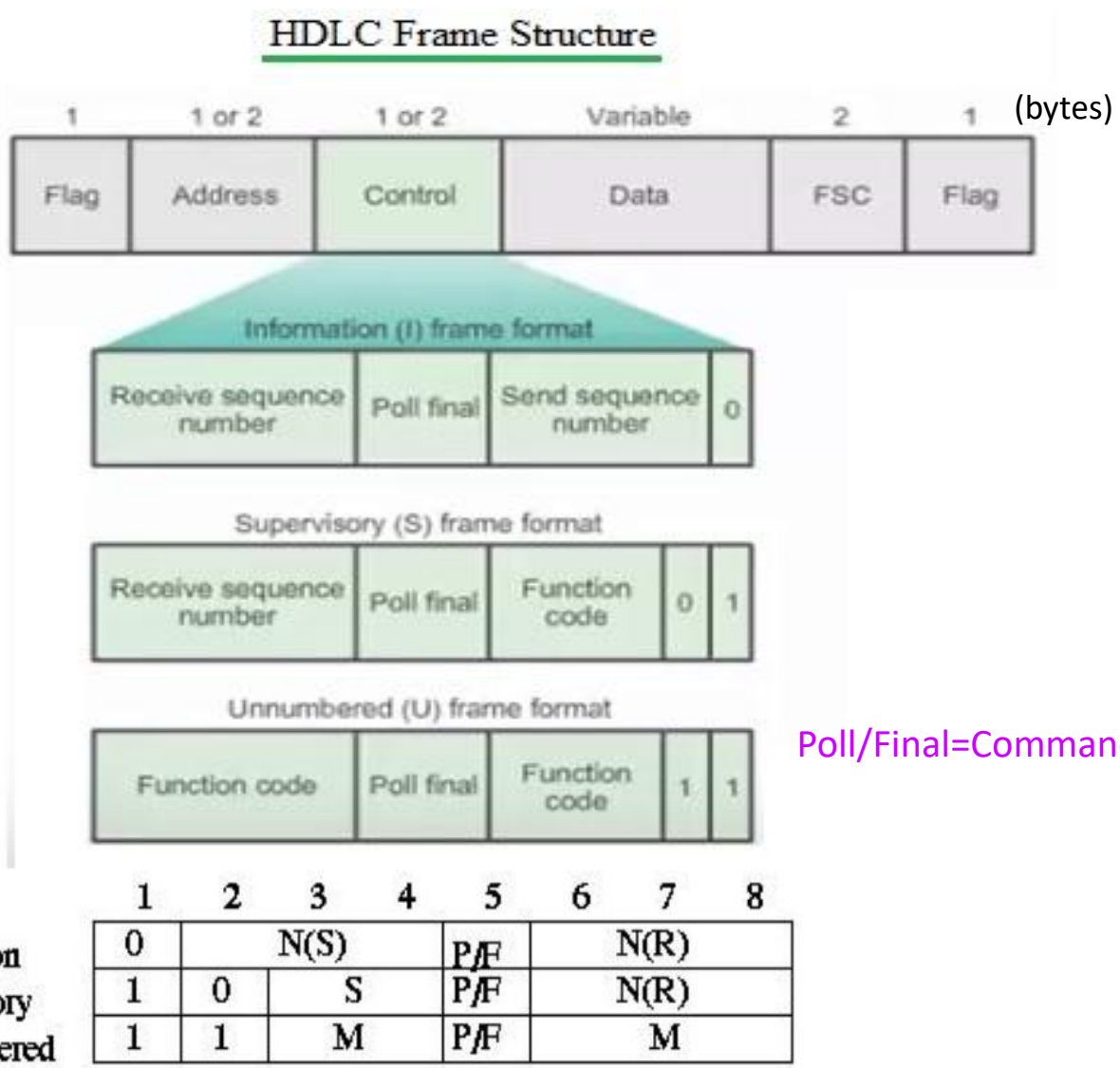
- Basic HDLC frame




- Flag
  - ✓ a signal to the receiver when a frame starts and ends
- Address
  - ✓ address of secondary station (destination in the view point of the primary)
- Control
  - ✓ type of the frame, sequence number, error tracking, etc.
- FCS
  - ✓ Frame Check Sequence used for error detection

Note Configuration Example:
https://networklessons.com/cisco/ccie-routing-switching-written/how-to-configure-hdlc-on-cisco-ios-router
http://howto-madkour.blogspot.com/2013/03/how-to-configure-hdlc-and-ppp-step-by.html
https://en.wikipedia.org/wiki/High-Level_Data_Link_Control

# HDLC Header




Poll/Final=Command/Response

- There are three different types of HDLC frame (specified by the control field)
  - ✓ Information frame (I-frame): carry information
  - ✓ Supervisory frame (S-frame): carry flow and error control information
  - ✓ Unnumbered frame (U-frame): reserved for system and link management

I: Information
S: Supervisory
U: Unnumbered

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| 0 | N(S) | | | P/F | N(R) | | |
| 1 | 0 | S | | P/F | N(R) | | |
| 1 | 1 | M | | P/F | M | | |

N(S) = Send sequence number
N(R) = Receive sequence number
S = Supervisory function bits
M = Unnumbered function bits
P/F = Poll/final bit

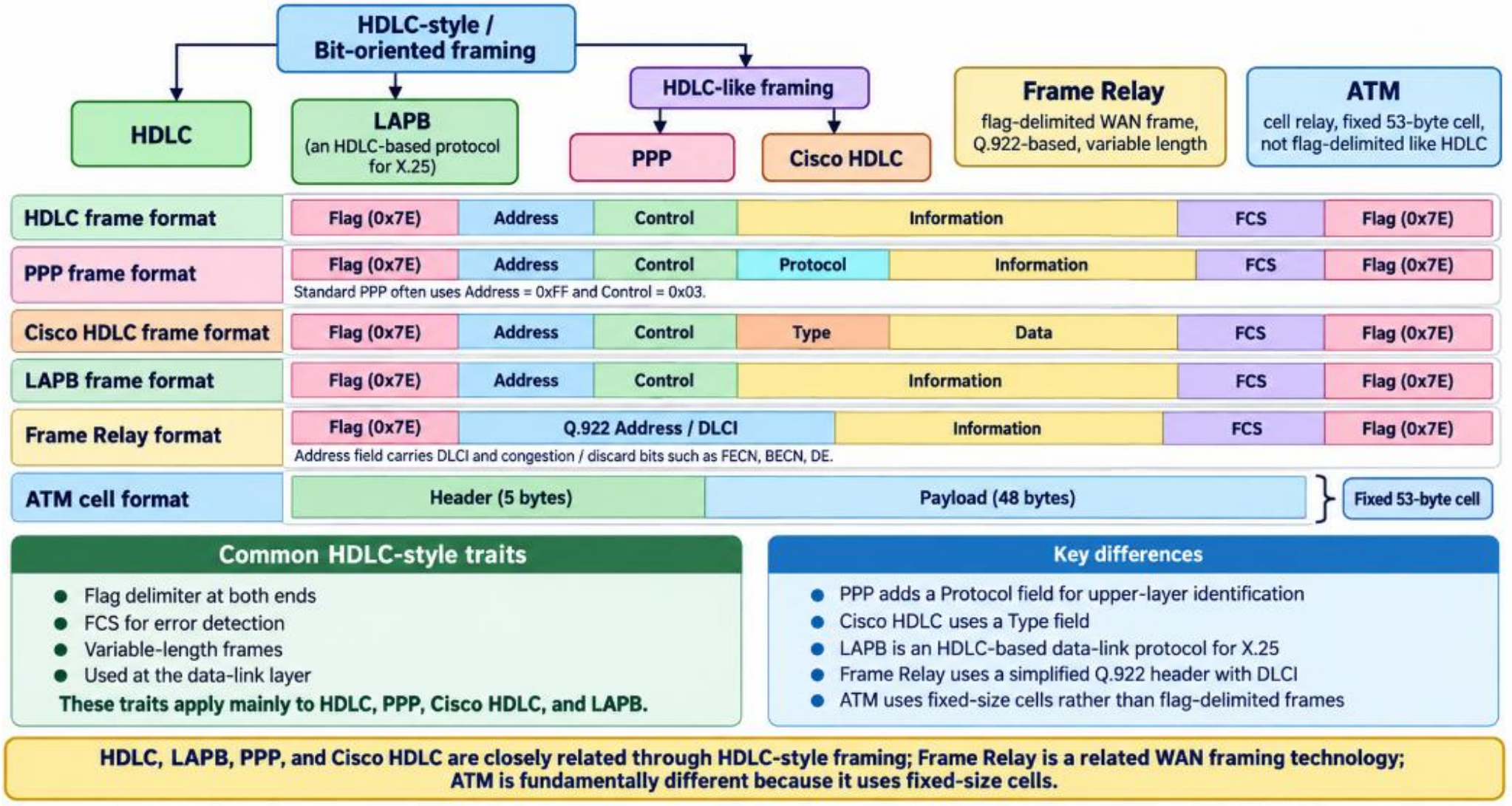
**Supervisory Function bits**
Receive Ready (RR, S bits=00)
REJect (REJ, S bits=01)
Receive Not Ready (RNR, S bits=10) – Possibly,
Selective REJect (SREJ, S bits=11)

**Packet tracer**




https://www.youtube.com/watch?v=ne6G1WvbFzg

# HDLC and Other Data Link Protocols



# Let’s have a Break !



# And do the Exercises Part 1





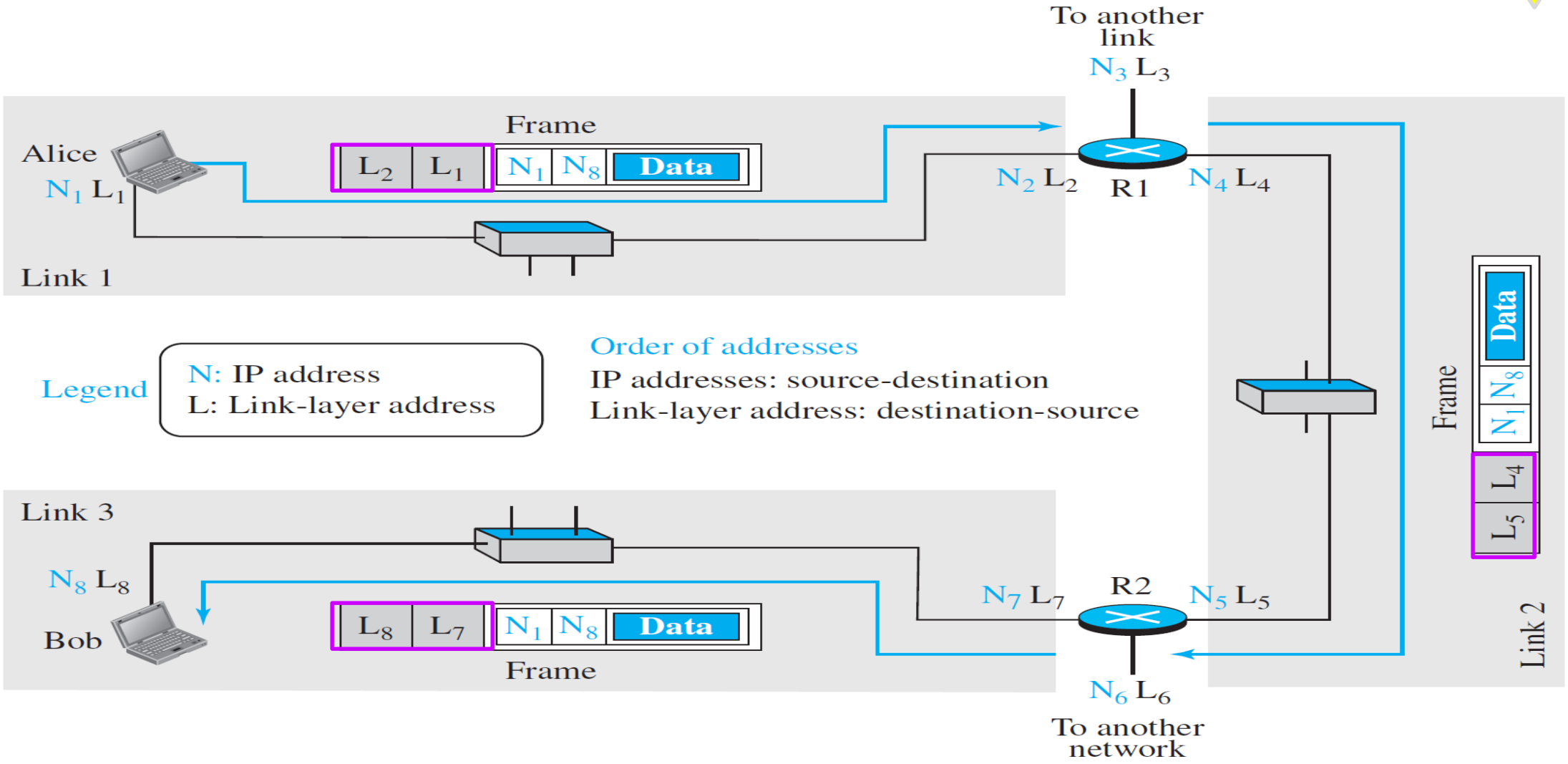




# Recap: Addressing

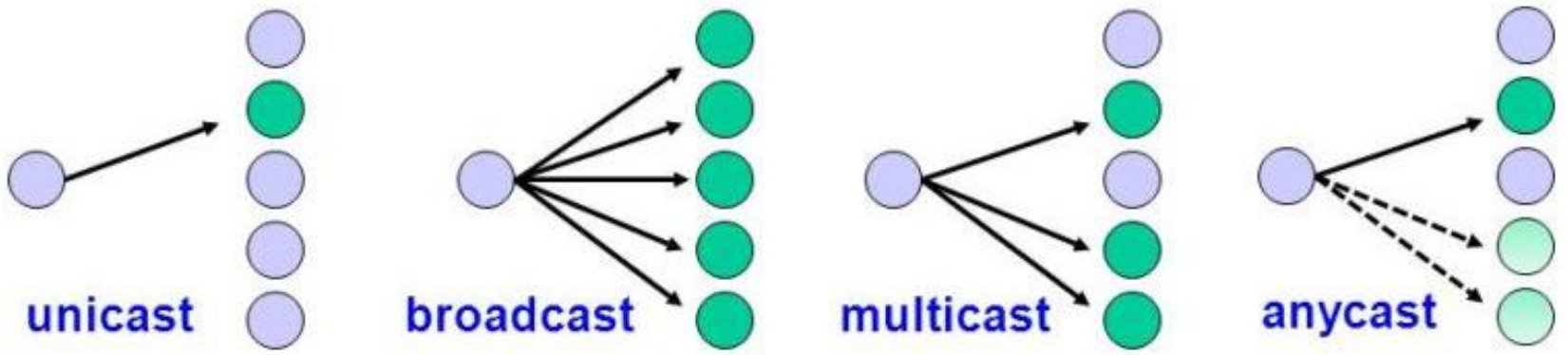
- In the case of having to communicate with many people in a network, a method of identifying a device is needed.
- Usually, it is called "**address**".
- There are 3 types of addresses used in the Internet
  - ✓Link layer address (MAC address/physical address)
  - ✓Internet protocol address (IP address)
  - ✓Application address (port number)

# Addressing


To another link
N3 L3
Alice
N1 L1
Frame
L2 L1 N1 N8 Data
N2 L2
R1
N4 L4
Link 1
Legend
N: IP address
L: Link-layer address
Order of addresses
IP addresses: source-destination
Link-layer address: destination-source
Frame
Data
N1 N8
L4
L5
Link 3
N8 L8
Bob
L8 L7 N1 N8 Data
Frame
N7 L7
R2
N5 L5
Link 2
N6 L6
To another network

# Communication Delivery Patterns (Casting)

- Casting describes how a sender delivers data to its intended receiver(s).
- The most important aspect is the type of communication or **casting** (sending data to audience). There are four types of castings.
  - ✓ Unicast: One-to-one
  - ✓ Anycast: One-to-one-of-many
  - ✓ Multicast: One-to-selected-many link
  - ✓ Broadcast: One-to-all (within a broadcast domain) link



# Types of MAC Destination Addresses

- ✓ Casting describes who should receive the data, while the destination address identifies that receiver or group at a particular layer.
- ✓ At the Data Link Layer, Ethernet supports unicast, multicast, and broadcast destination MAC addresses. Anycast is normally implemented at the Network Layer using IP addressing and routing.

- Unicast address
  - ✓ A unique address specifying a node

    Applications: from PC to the gateway NIC address, ARP reply, DHCP ack, DHCP offer

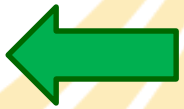
- Multicast address
  - ✓ One address links to several members

    Applications: Streaming (IPTV), routing protocols (OSPF, EIGRP), IPv6 neighbor discovery, multicast DNS (mDNS)
    e.g. multicast IPv4: 224.0.0.251 maps to MAC 01:00:5E:00:00:FB
- Broadcast address
  - ✓ Refer to everyone FF:FF:FF:FF:FF:FF

    Applications: ARP request, DHCP request, DHCP discover

# Obtaining an Address

- There are 3 types of addresses mainly used in computer data communication
  - **Physical Address or MAC Address**
    - ✓ The address used in the data link layer. It is usually used to identify a device within a local area network.
  - **Internet Protocol (IP) Address**
    - ✓ The address used in the network layer. It is usually used to identify a device in a global network.
  - **Application Address (Port Number)**
    - ✓ The address used in the application layer. It is usually used to identify a program used for reading/interpreting the received data.
- **How does a computer determine its own address? Who provide the addresses?**



# Checking MAC Address

## For Mobile Devices

- **iOS:**
  1. Open the Settings app.
  2. Tap on General, then About.
  3. Scroll down and find the Wi-Fi Address, which is the device's MAC address.

- **Android:**
  1. Open the Settings app.
  2. Navigate to About phone or About device.
  3. Tap on Status or Hardware information.
  4. Look for the Wi-Fi MAC address.

## For Computers



- **macOS:**
  1. Open the Apple Menu (□).
  2. Select System Preferences.
  3. Click on Network.
  4. Select the Wi-Fi network.
  5. Click the Advanced button, then select the Hardware tab. The Wi-Fi Address is your MAC address.

- **Linux:**
  1. Open a terminal window.
  2. Type `ip link show` or `ifconfig -a` and press Enter.
  3. Look for the MAC address associated with your network interface (e.g., eth0), often shown as HWaddr or ether.

### Windows

1. Open the Command Prompt.
2. Type the command `ipconfig /all` and press Enter.
3. Look for the Physical Address heading to find the MAC address for your network adapter.

# Checking IP Address

### iOS (iPhone/iPad)

1. Open the Settings app.
2. Tap on Wi-Fi.
3. Tap the "i" icon (information icon) next to the name of the Wi-Fi network you are connected to.
4. Look for the "IP Address" or "IPv4 Address" listed.

### Android

1. Open the Settings app.
2. Navigate to About phone or About device.
3. Select Status or IP address to find your device's IP address.

### macOS



1. Click the Apple menu (□) in the top-left corner.
2. Select System Settings (or System Preferences).
3. Click Network in the sidebar.
4. Select Wi-Fi from the list of services.
5. Click the "Details..." button next to your connected network.
6. Select the TCP/IP tab to find the IPv4 Address.

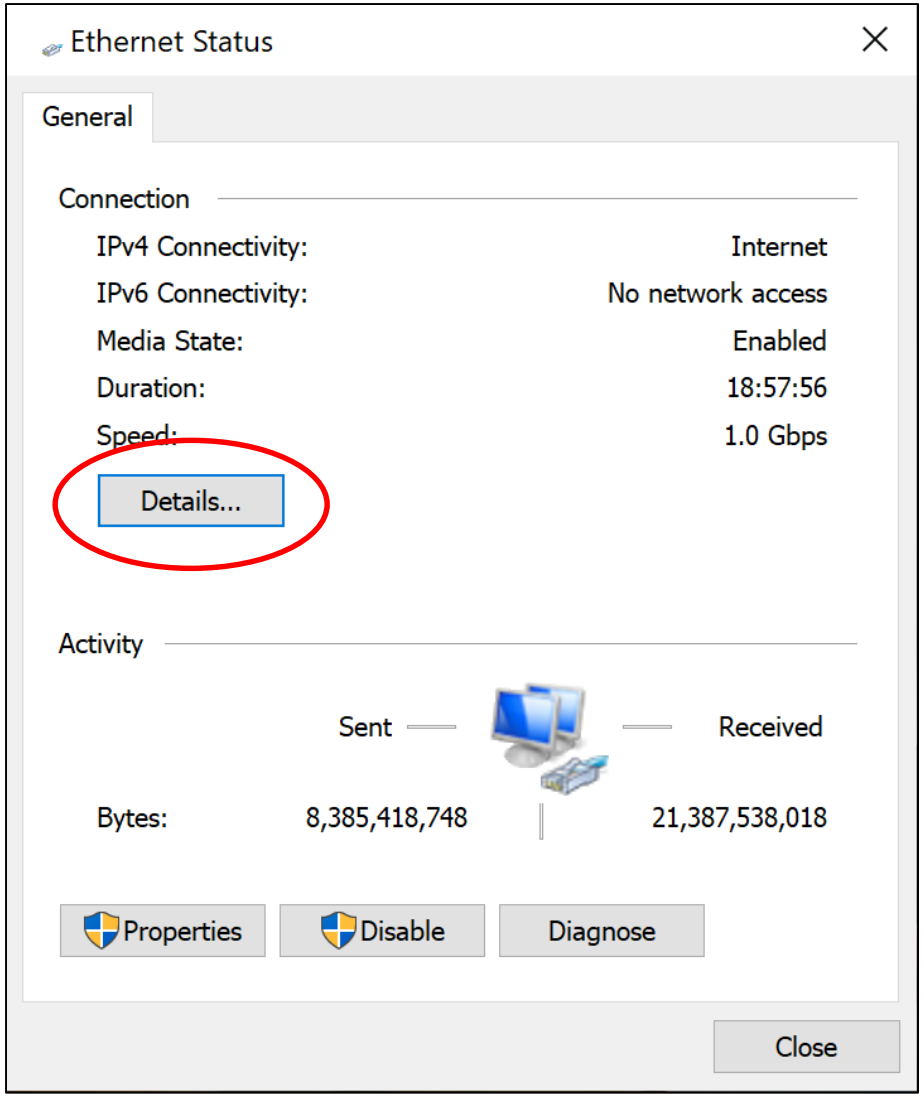
### Windows

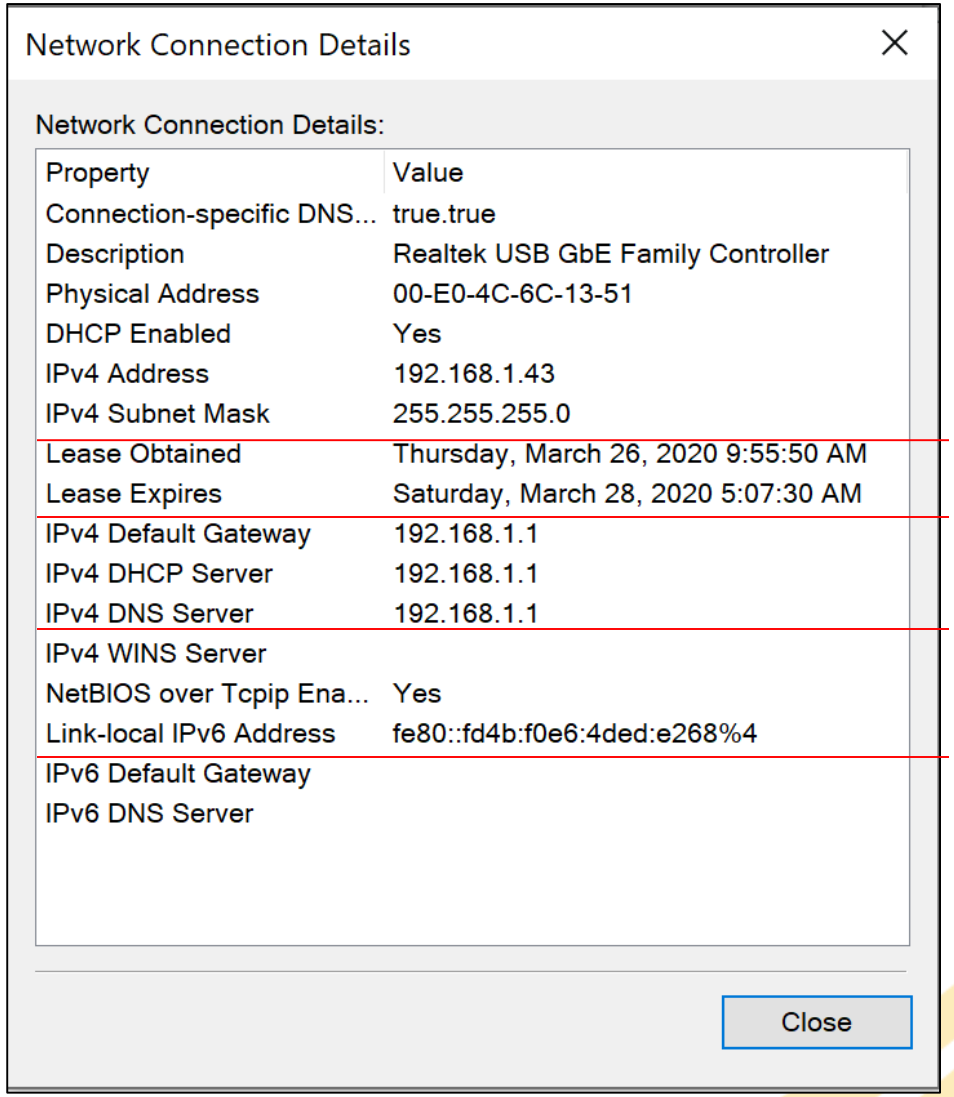
1. Open the Start Menu and type `cmd` or `Command Prompt`.
2. Open the Command Prompt window.
3. Type the command `ipconfig /all` and press Enter.
4. Look for the IPv4 Address under your active network adapter.

### Linux

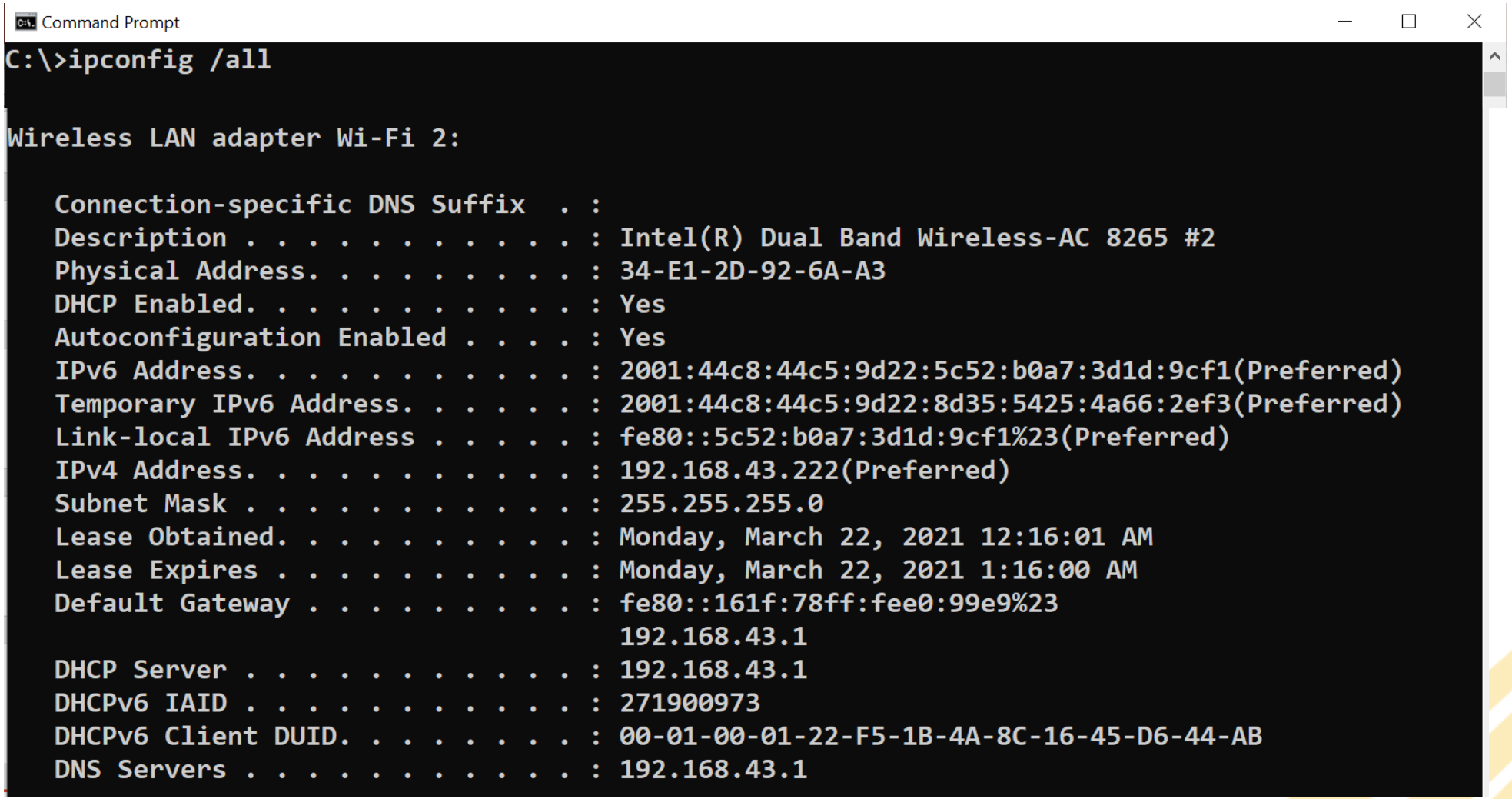
1. Open a terminal window.
2. Type either `ip a` or `ifconfig` and press Enter.
3. Look for the IP address listed for your network interface (e.g., eth0 or enpXsX).

# Checking Computer's Address



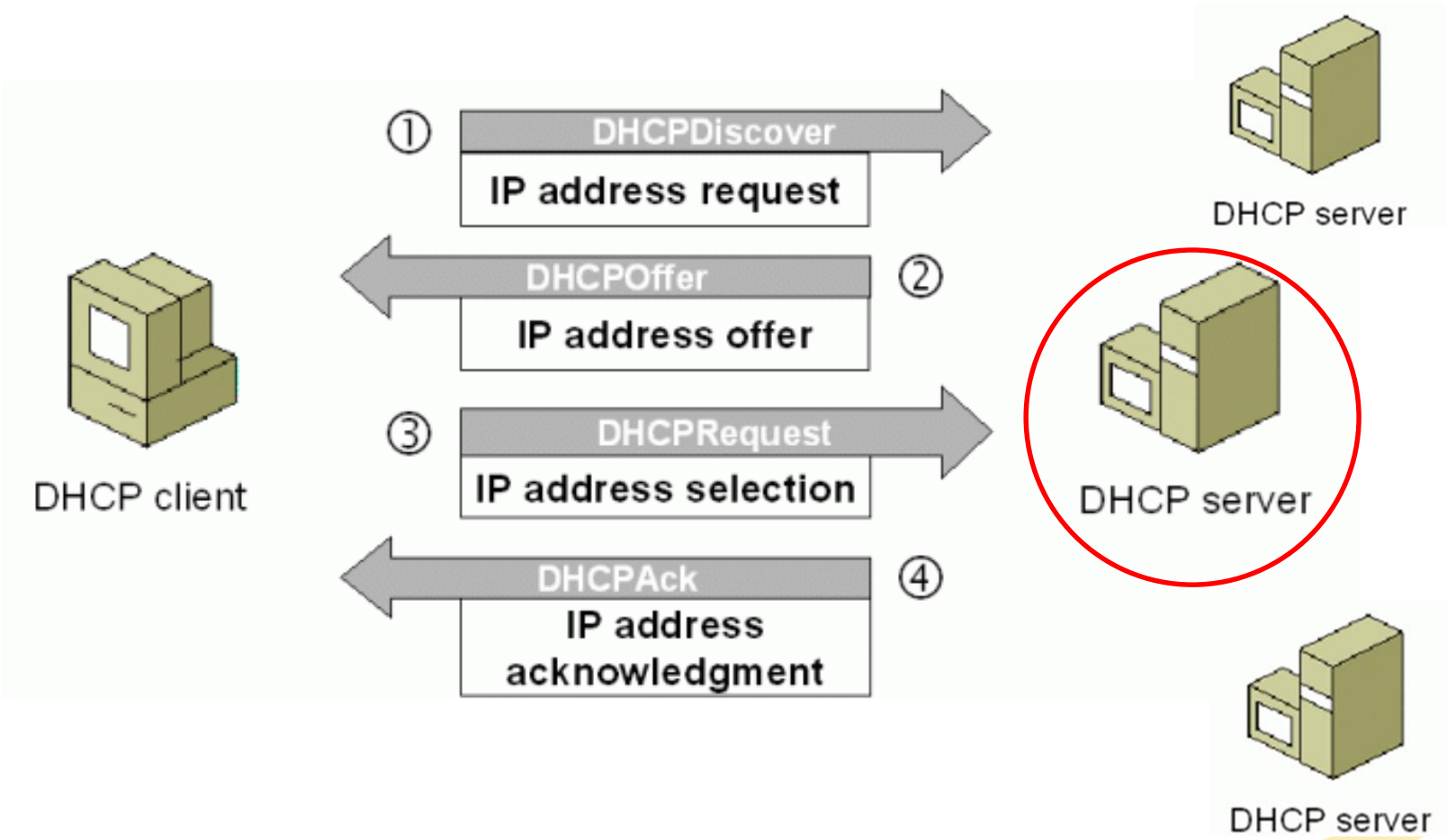
# ipconfig /all



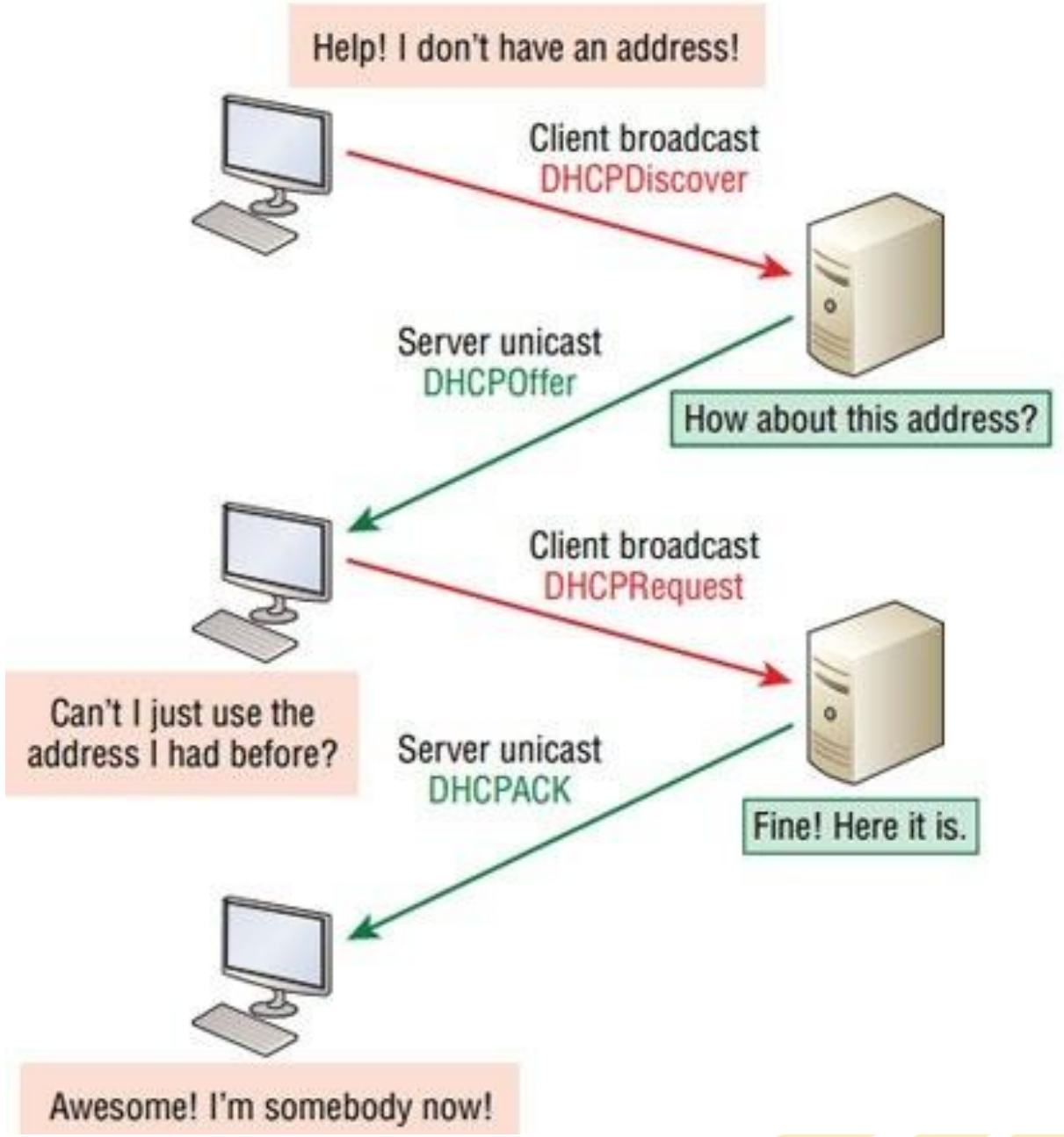
# DHCP (RFC 2131 / RFC 2132)

- **Dynamic Host Configuration Protocol (DHCP)** is a protocol used for a new machine to obtain IP address, subnet mask, gateway, and other network configuration information from the network it is connected to and is developed from BOOTP (Bootstrap Protocol:RFC951/1048)

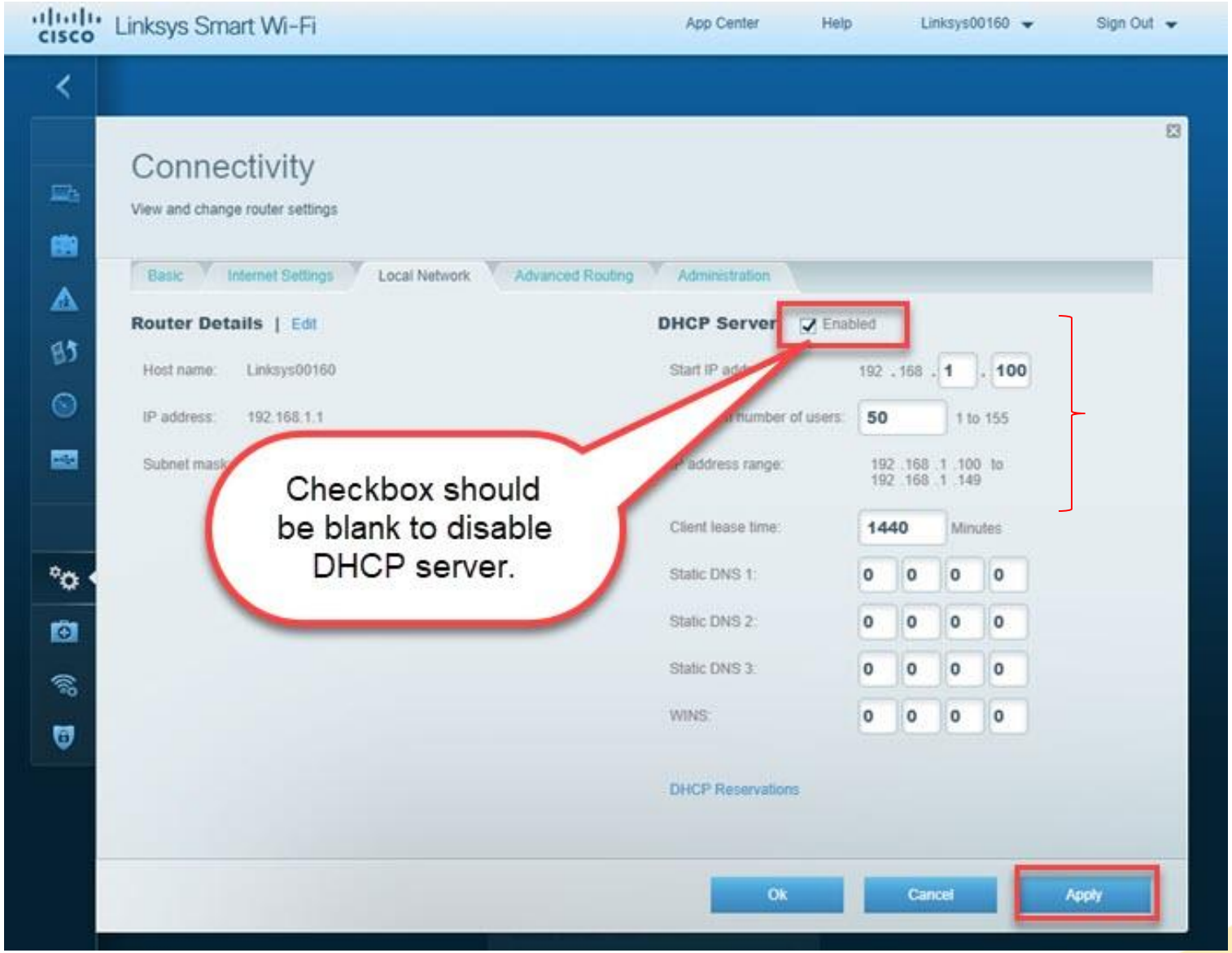
"DORA"



# DHCP Request in Human Context

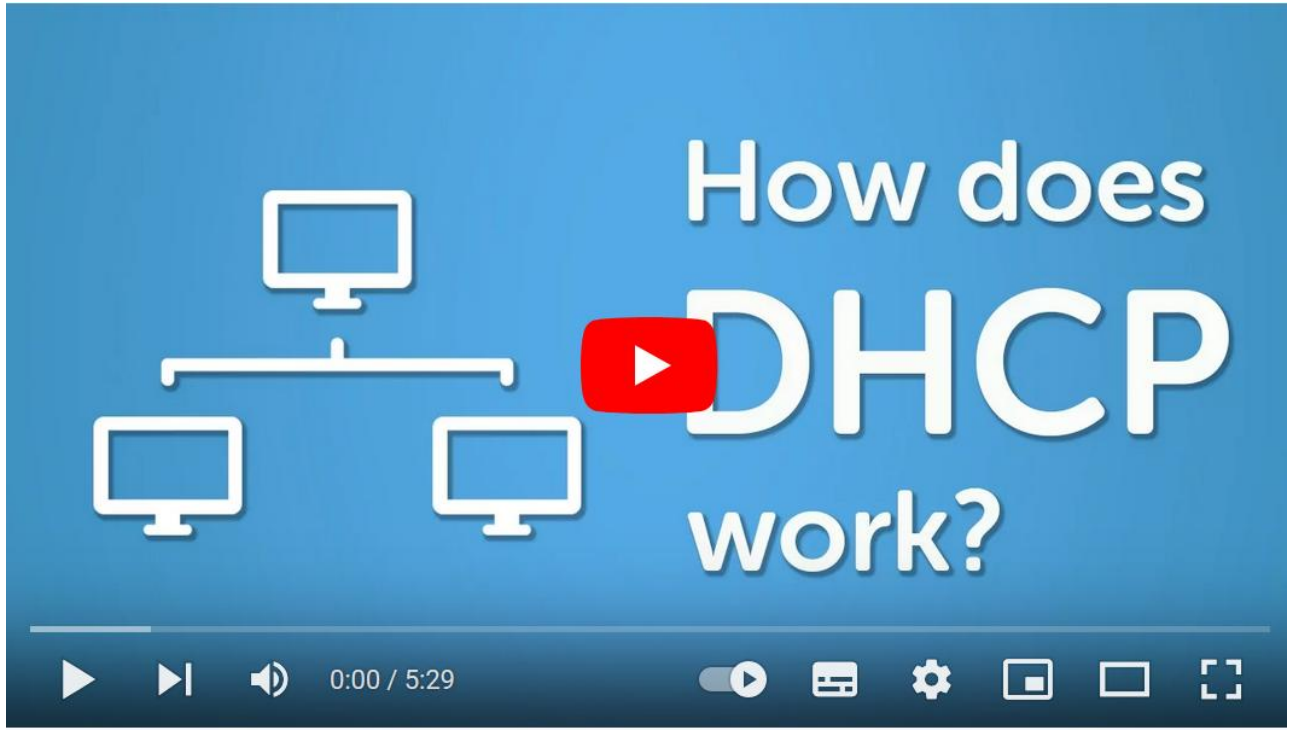


# Configure DHCP Server



# DHCP in Details

- https://www.youtube.com/watch?v=S43CFcpOZSI

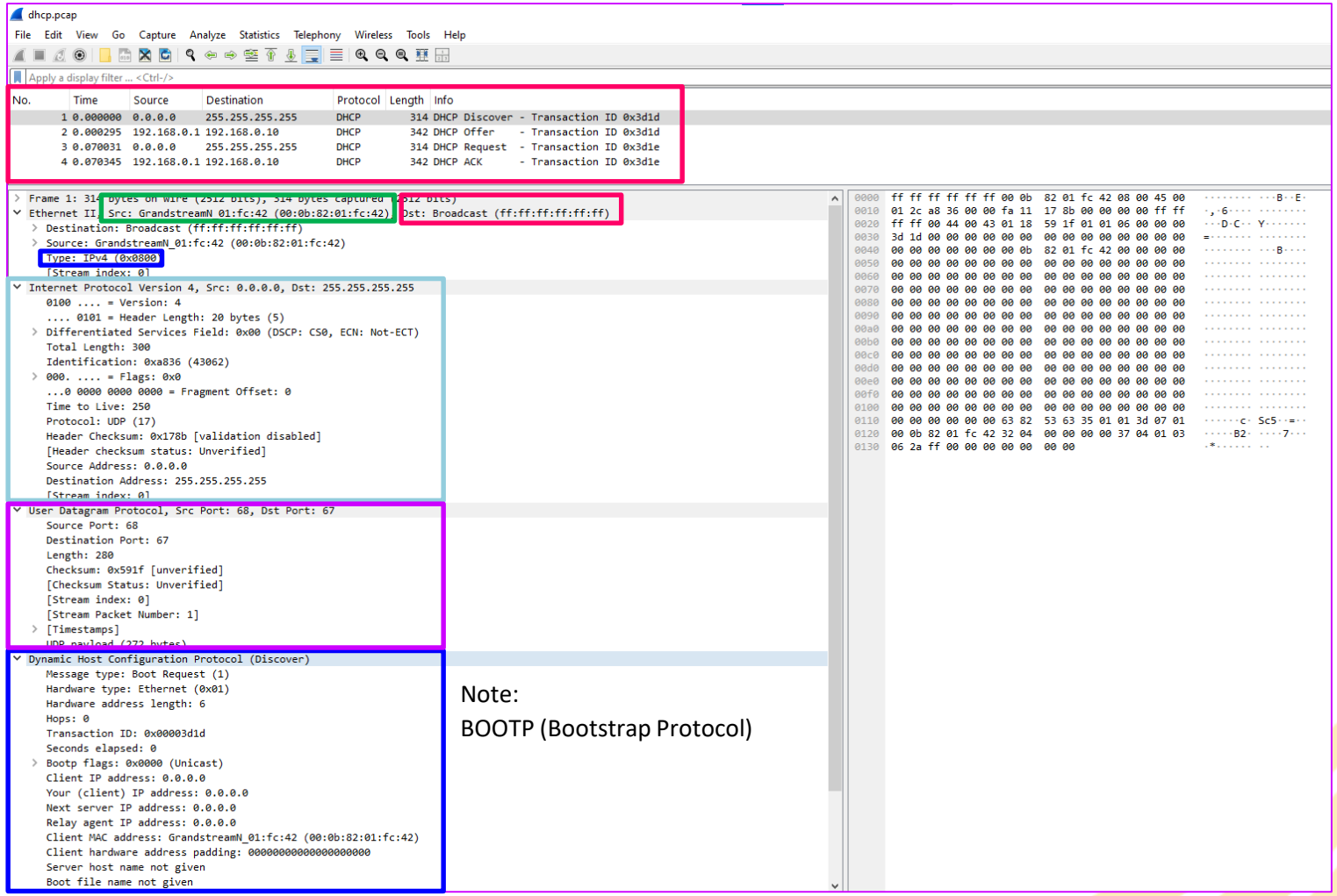



- For more information,
  - https://www.netmanias.com/en/post/techdocs/5998/dhcp-network-protocol/understanding-the-basic-operations-of-dhcp

# Wireshark

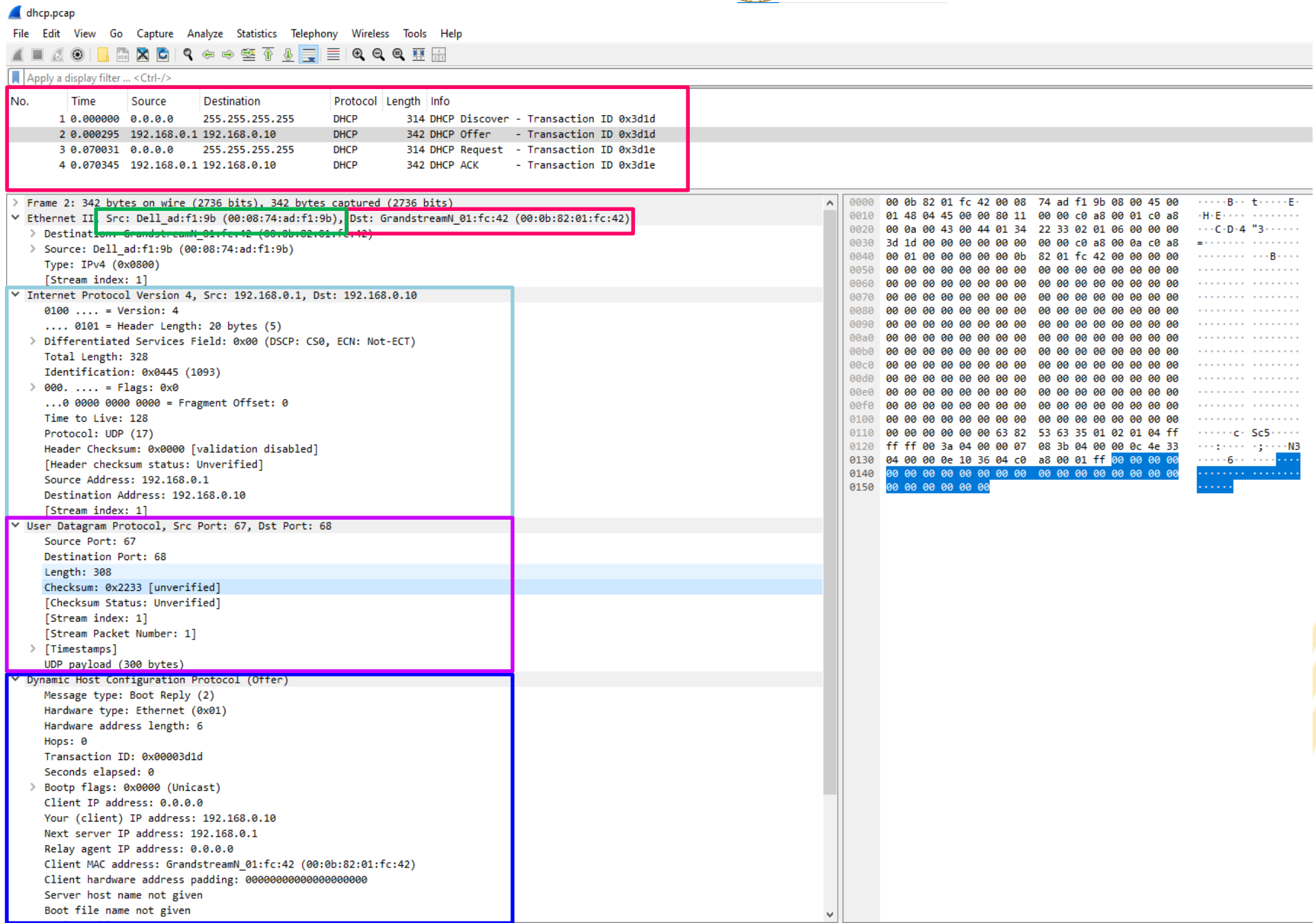
- https://www.wireshark.org/download.html
- https://wiki.wireshark.org/samplecaptures
- https://wiki.wireshark.org/Development/PcapNg

## DHCP Discover



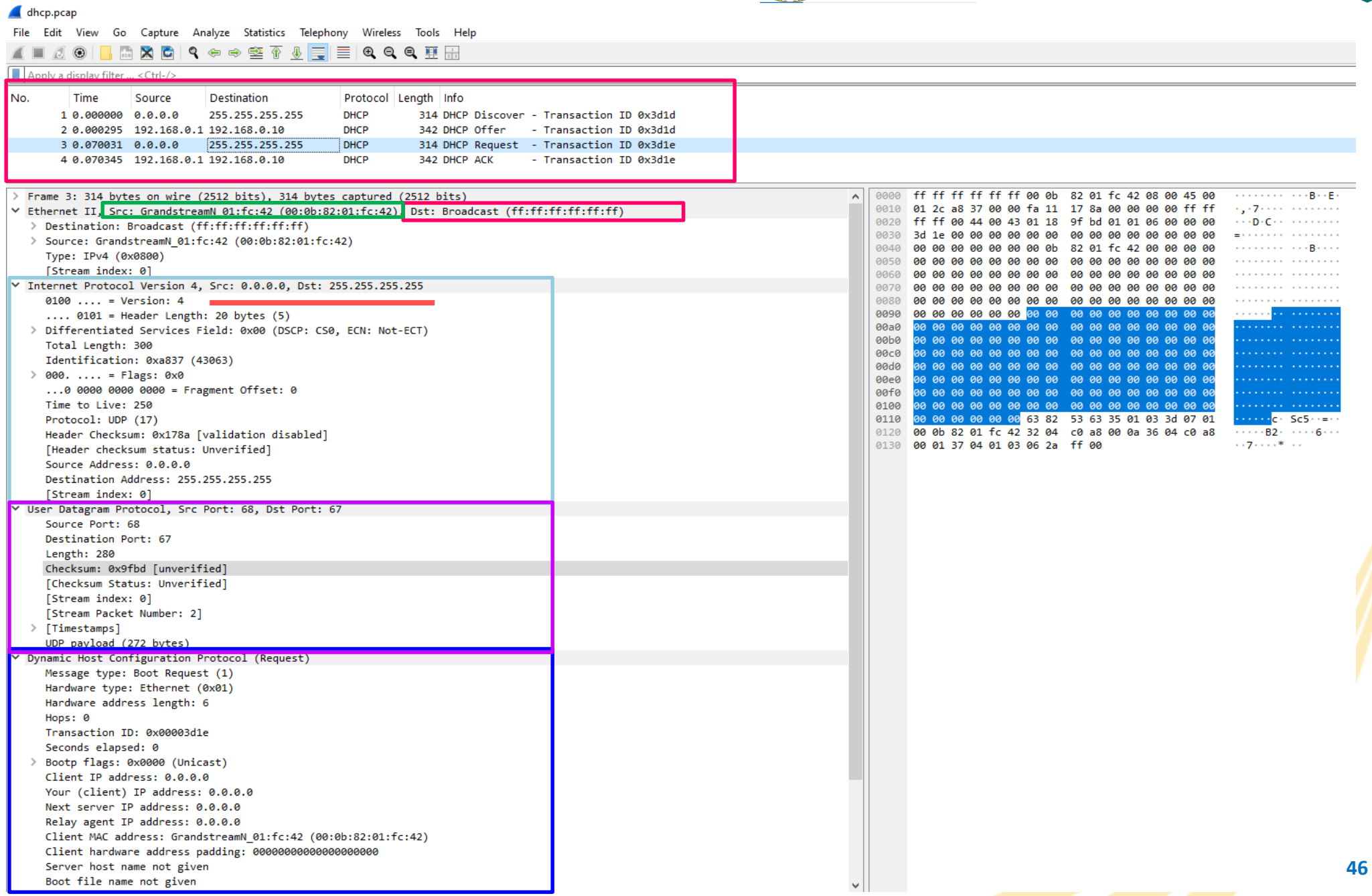
# Wireshark

## DHCP Offer



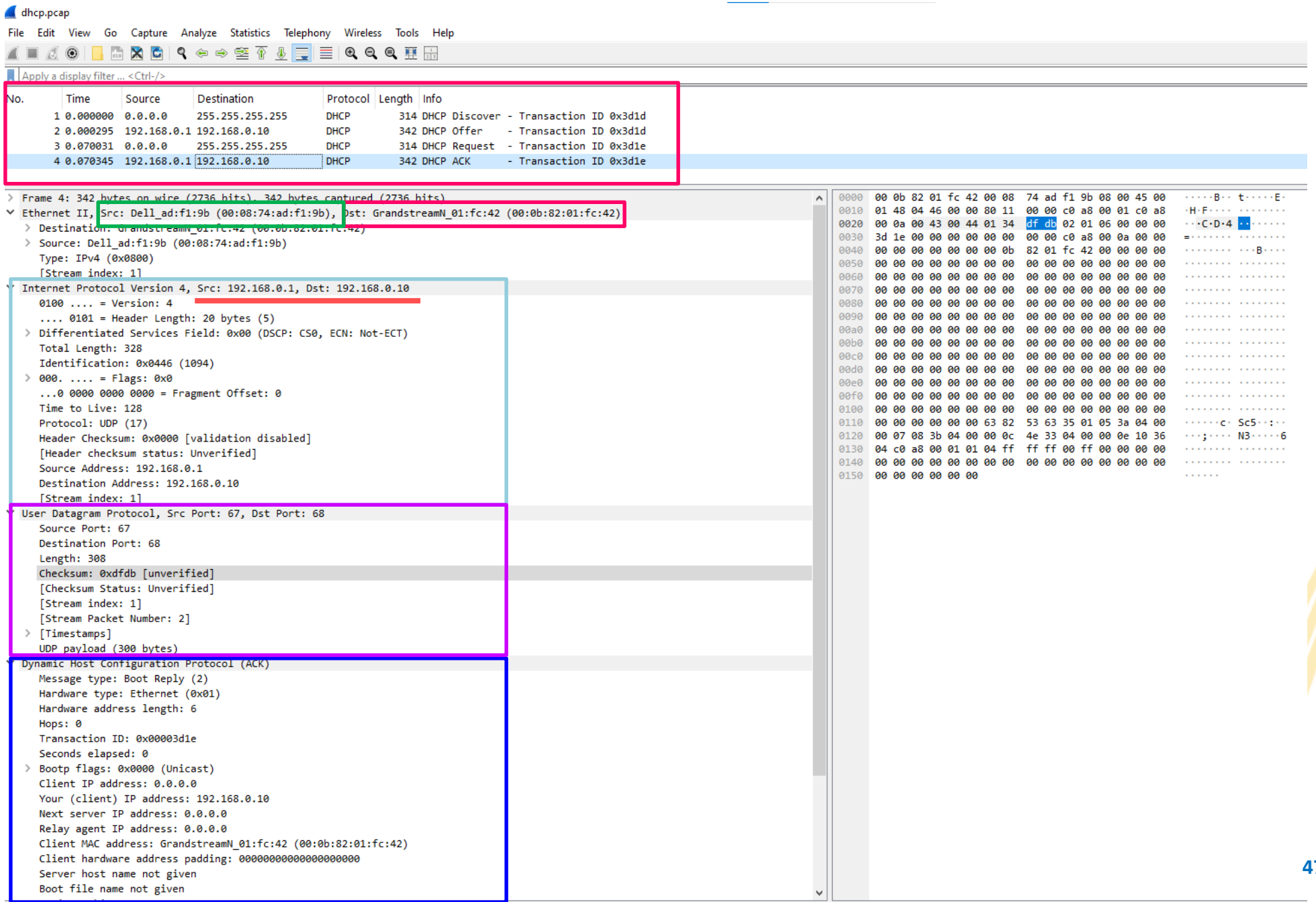
# Wireshark

## DHCP Request



# Wireshark

# DHCP ACK



# Wireshark

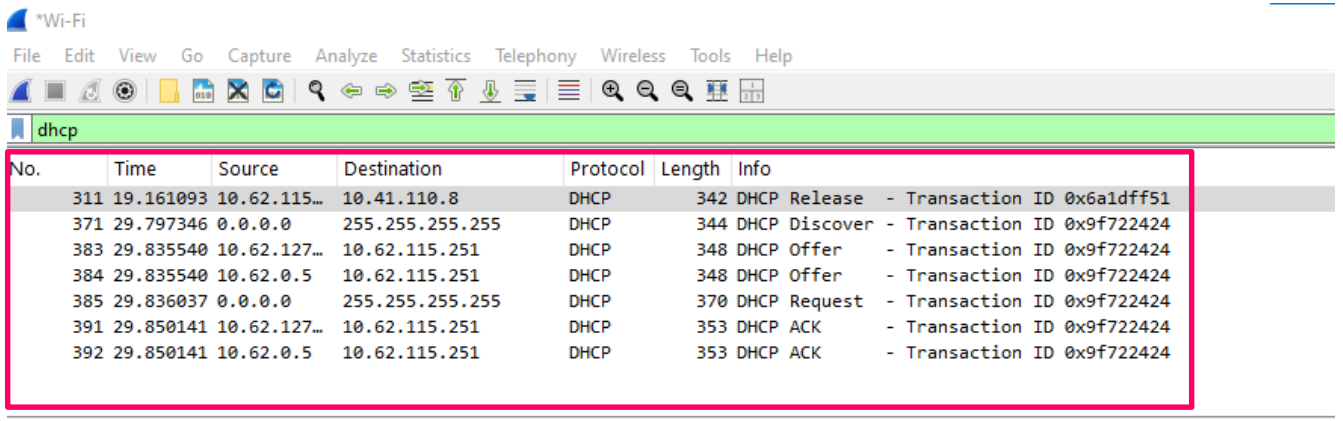
Capture DHCP

```
ipconfig /release
```

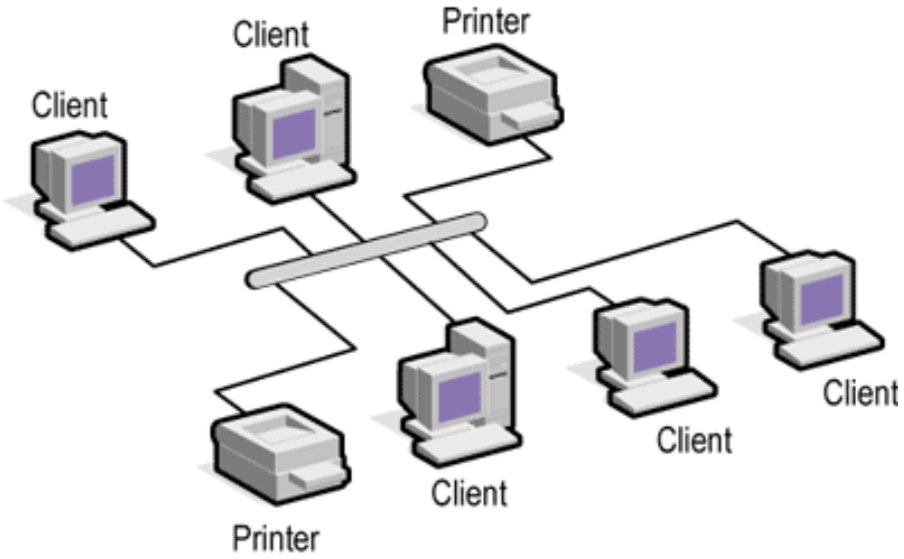
```
sudo dhcplient -r
```

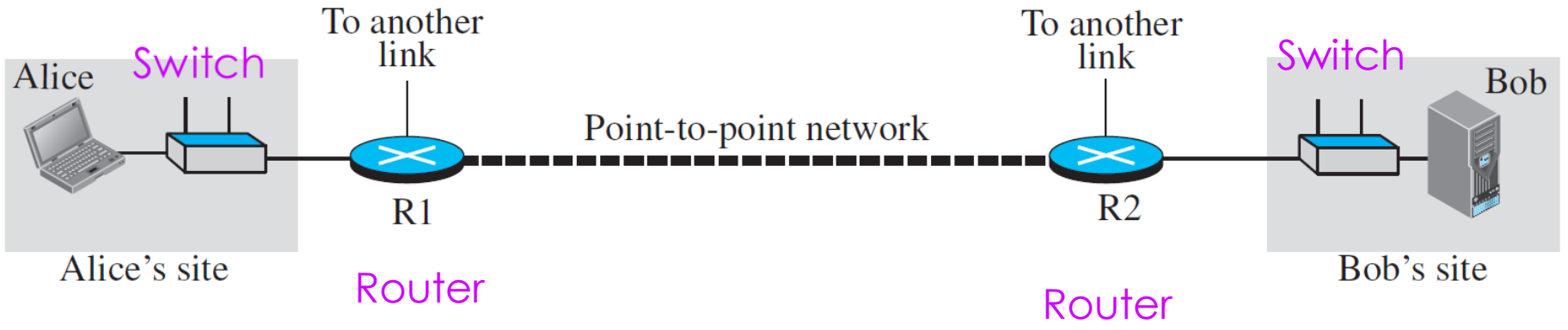
```
sudo nmcli device disconnect eth0
sudo nmcli device connect eth0

sudo networkctl renew eth0
sudo networkctl release eth0
```

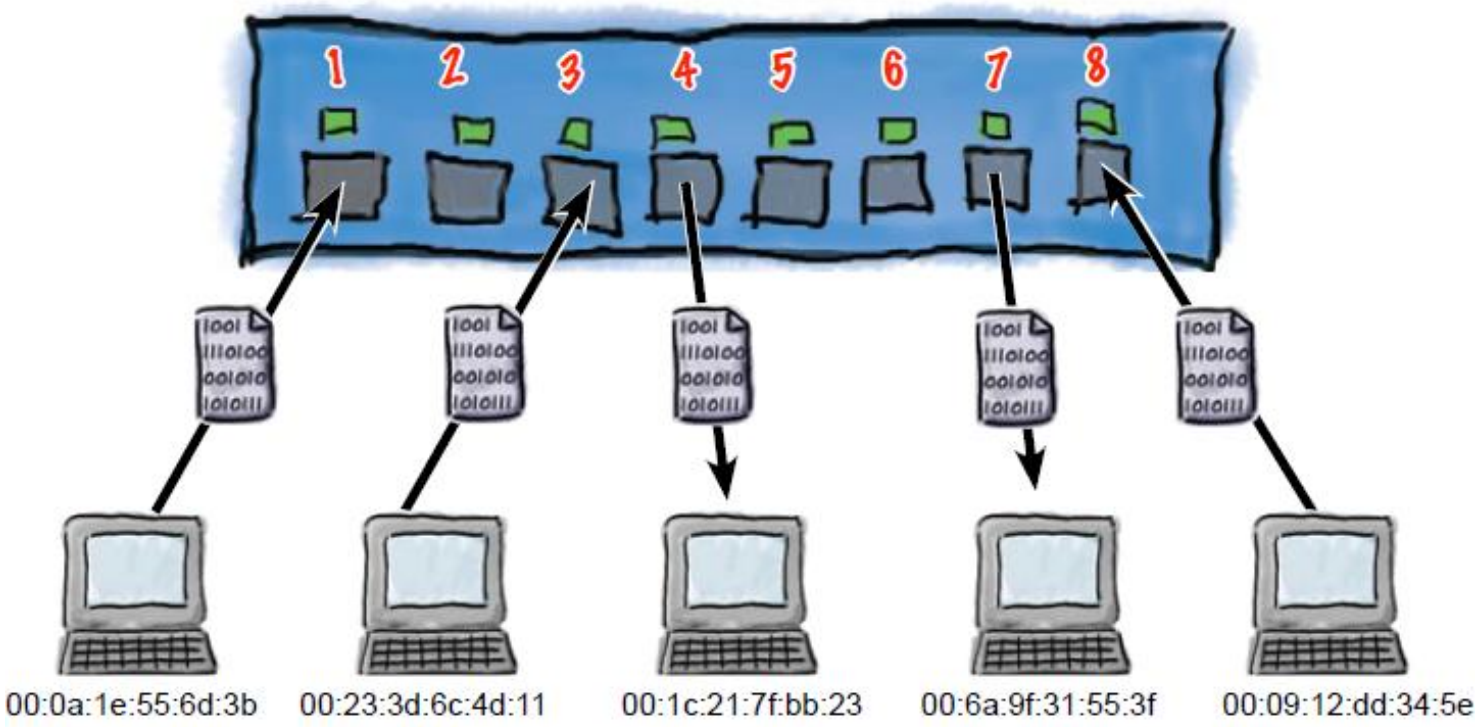


# Example of LAN/WAN Communication



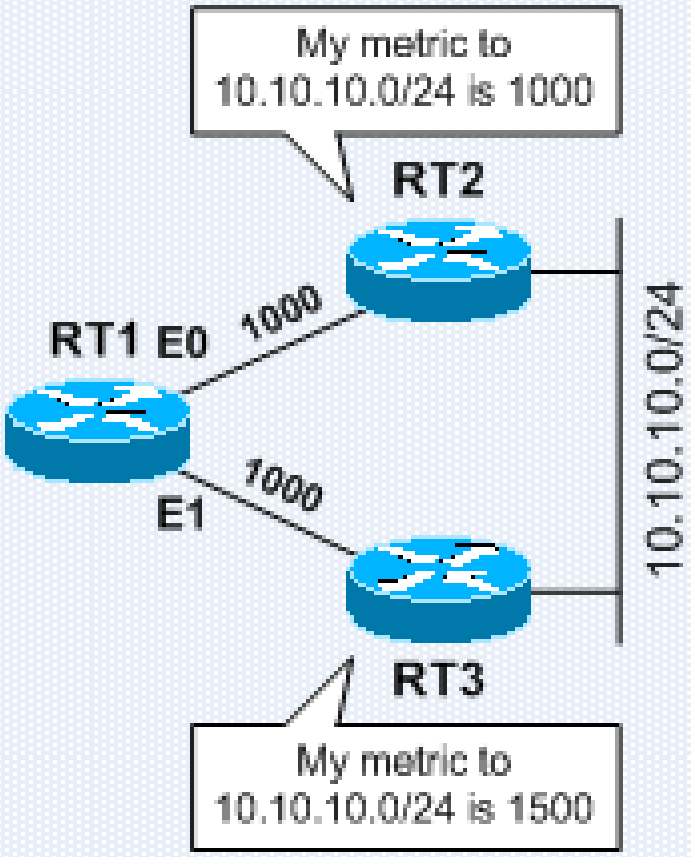



# Network Equipment: Switches (Layer 2)




| MAC address | Port |
|---|---|
| 00:0a:1e:55:6d:3b | 1 |
| 00:23:3d:6c:4d:11 | 3 |
| 00:1c:21:7f:bb:23 | 4 |
| 00:6a:9f:31:55:3f | 7 |
| 00:09:12:dd:34:5e | 8 |

# Network Equipment: Router (Layer 3)




**IP EIGRP Neighbor Table**

| Next-Hop Router | Interface |
|---|---|
| RT2 | E0 |
| RT3 | E1 |

Routing table

RT1

**IP EIGRP Topology Table**

| Network | Feasible Distance | Advertised Distance | EIGRP Neighbor |
|---|---|---|---|
| 10.10.10.0/24 | 2000 | 1000 | RT2 (E0) |
| 10.10.10.0/24 | 2500 | 1500 | RT3 (E1) |

**IP Routing Table**

| Network | Metric (FD) | Outbound Interface | Next-Hop |
|---|---|---|---|
| 10.10.10.0/24 | 2000 | E0 | RT2 |

# Network Equipment: Switch vs Router

| Switching | Routing |
|---|---|
| Layer 2 Switches perform Switching | Layer 3 devices like Router perform Routing |
| Switching will be faster as switch uses ASIC technology | Routing will be slower as it is software based. |
| Switching is done at layer 2 of OSI Model | Routing is done at layer 3 of OSI Model |
| If the destination is not known to switch it will broadcast the frame. | If the destination is not known to router it will drop the packet. |
| Switching is done in same broadcast domain. | Routing is done in different networks. |
| Switching is done by using MAC address. | Routing is done by using IP address. |
| Protocol data unit at layer 2 is frame | Protocol data unit at layer 3 is packet |

Switch



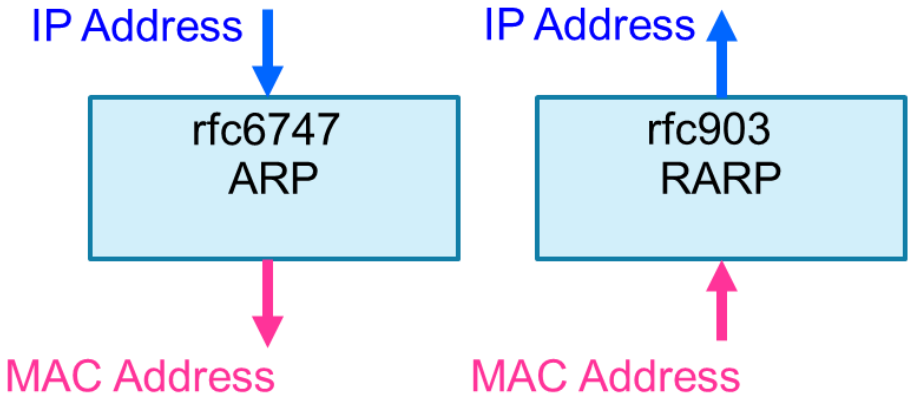
Router



Note:
ASIC stands for Application-specific integrated circuit. An ASIC is similar in theory to an FPGA, with the exception that it is fabricated as a custom circuit.

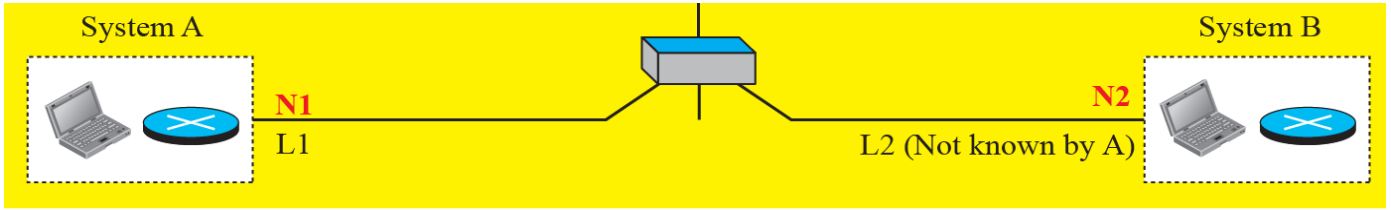
# Address Resolution Protocol (ARP)

- ARP is needed when a station or a node needs to know the link-layer address of a node.
- The requesting station will issue a broadcast request.
- Anyone whose address is mentioned in the broadcast request frame will respond by sending the corresponding MAC address of the node.

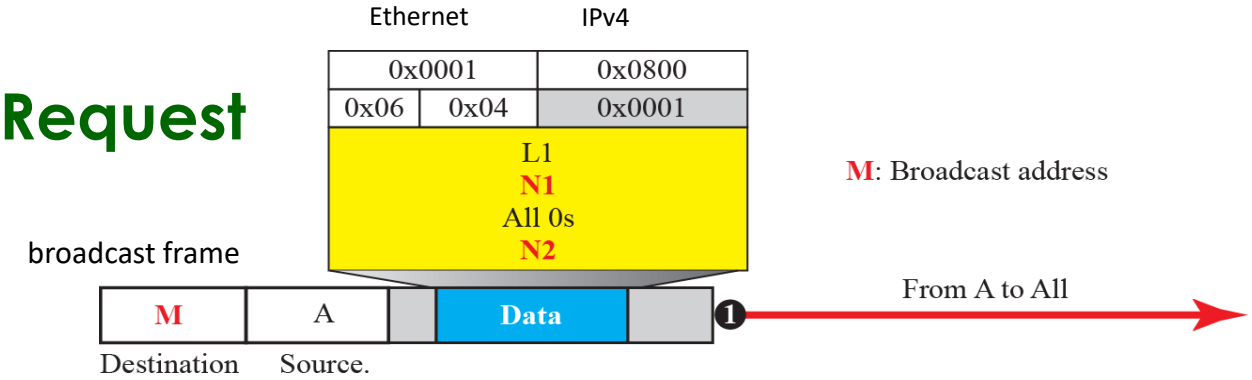



rfc 826: Ethernet ARP
rfc1931: Dynamic RARP

# ARP Message Types: ARP Request/ARP Reply

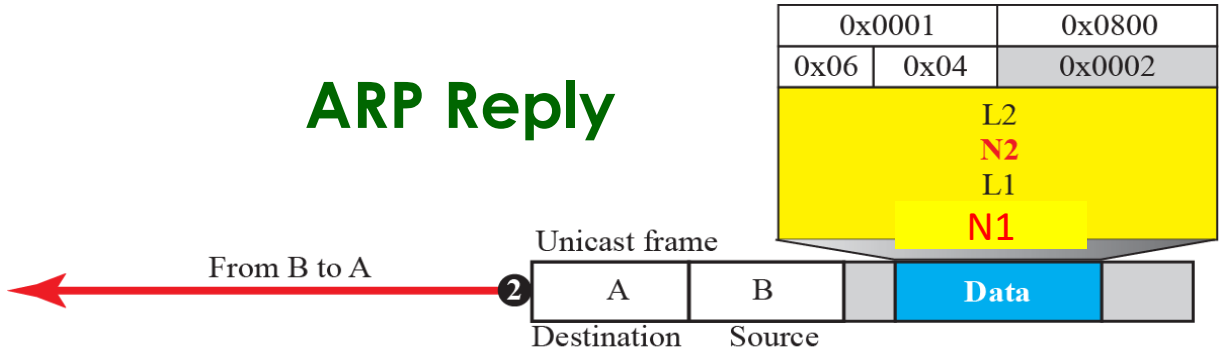



## ARP Request

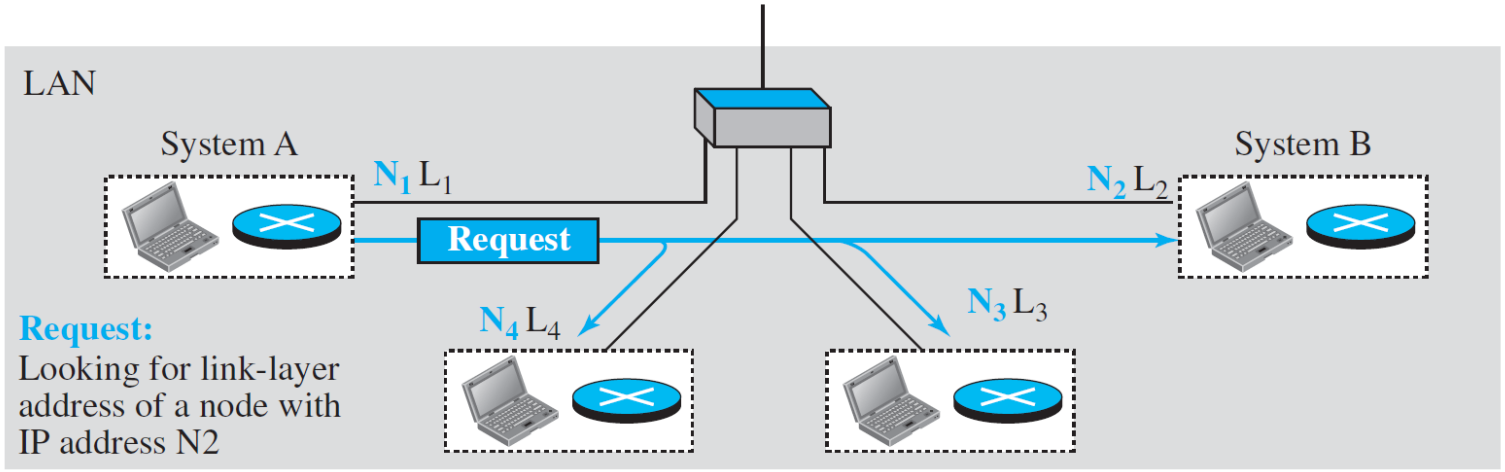



| Hardware Type | | Protocol Type |
|---|---|---|
| Hardware length | Protocol length | Operation Request 1, Reply 2 |
| Sender hardware address (For example, 6 bytes for Ethernet) | | |
| Sender protocol address (For example, 4 bytes for IP) | | |
| Target hardware address (For example, 6 bytes for Ethernet) (It is not filled in a request) | | |
| Target protocol address (For example, 4 bytes for IP) | | |

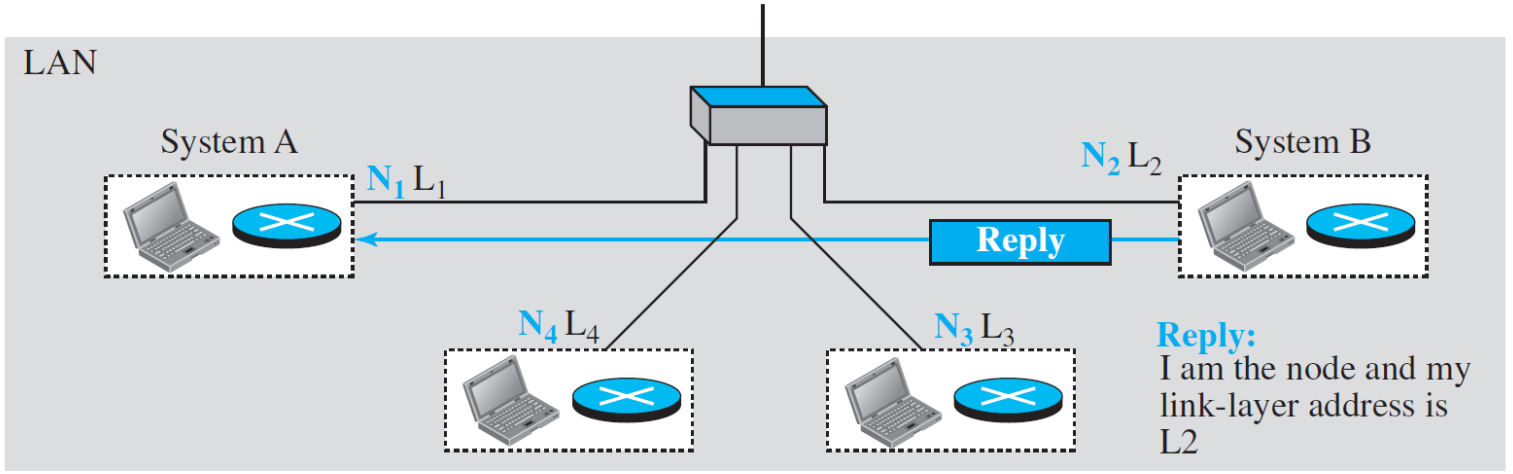
## ARP Reply



# ARP Operations in the Same Network




a. ARP request is broadcast




b. ARP reply is unicast

# arp -a

Command Prompt

```
C:\>arp -a

Interface: 192.168.56.1 --- 0x4
  Internet Address      Physical Address      Type
  192.168.56.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.192.152.143       01-00-5e-40-98-8f     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
```

# arp -d

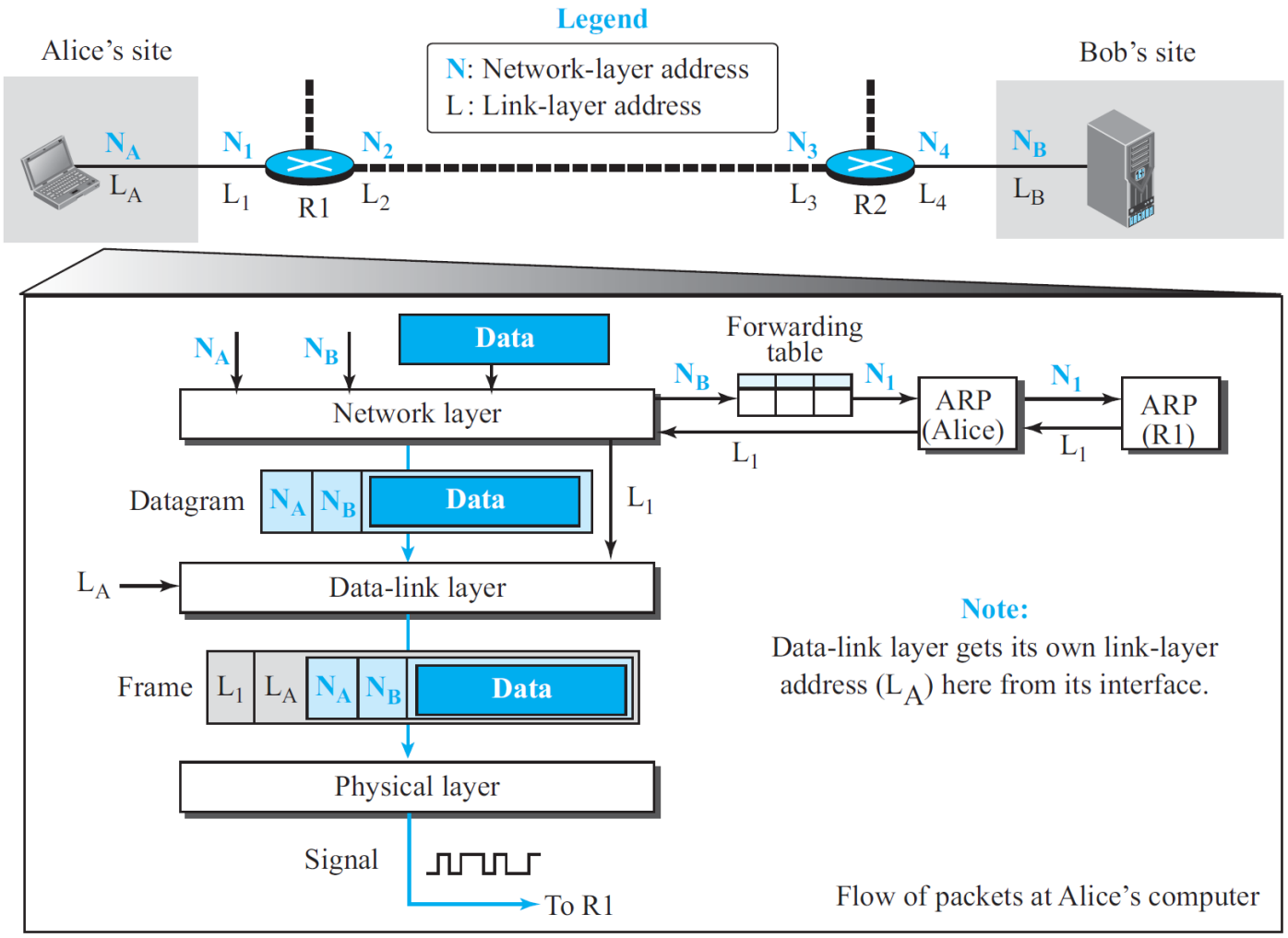
```
netsh interface ip delete arpcache
```

delete (by Admin)

```
┌──(kali㉿kali)-[~]
└─$ arp -a
? (10.0.2.2) at 52:54:00:12:35:00 [ether] on eth0
? (10.0.2.13) at 08:00:27:51:5b:ed [ether] on eth0
```

# ARP Operations in the Different Networks Internet Communication Example (Step 1)

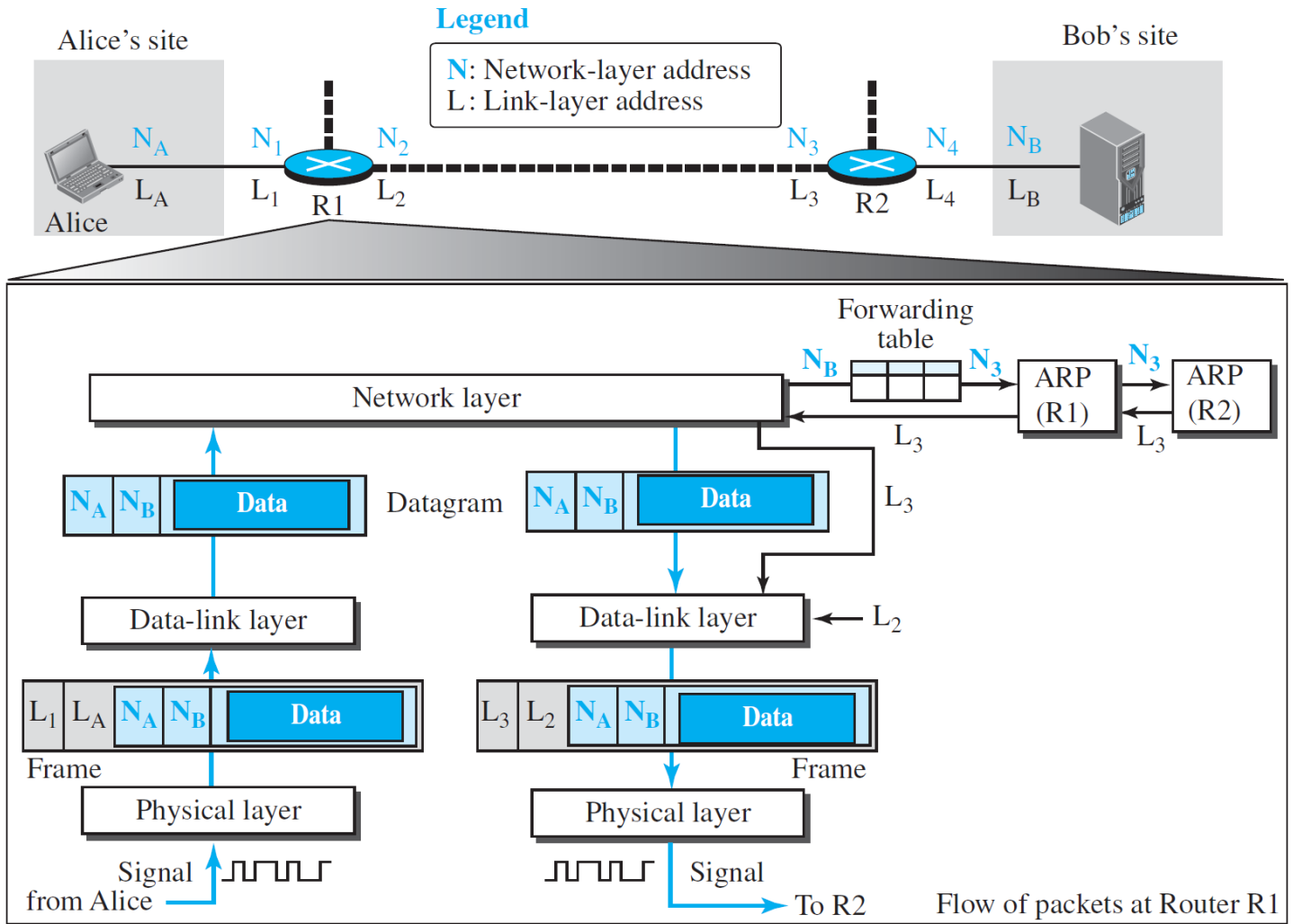
The sender (Alice) sends data to a router gateway. Even though the sender knows the gateway IP address, the sender needs to use ARP to get the gateway's MAC address in order to send the frame in the data link layer.




- Alice: Sender ($N_A$)
  Bob: Destination ($N_B$)
- Alice sends data to a router gateway ($R_1$).
- Even though Alice knows the gateway IP address ($N_1$),
- Alice needs to use ARP to get the gateway ($N_1$)'s MAC address ($L_1$) in order to send the frame in the data link layer.

# ARP Operations in the Different Networks Internet Communication Example (Step 2)

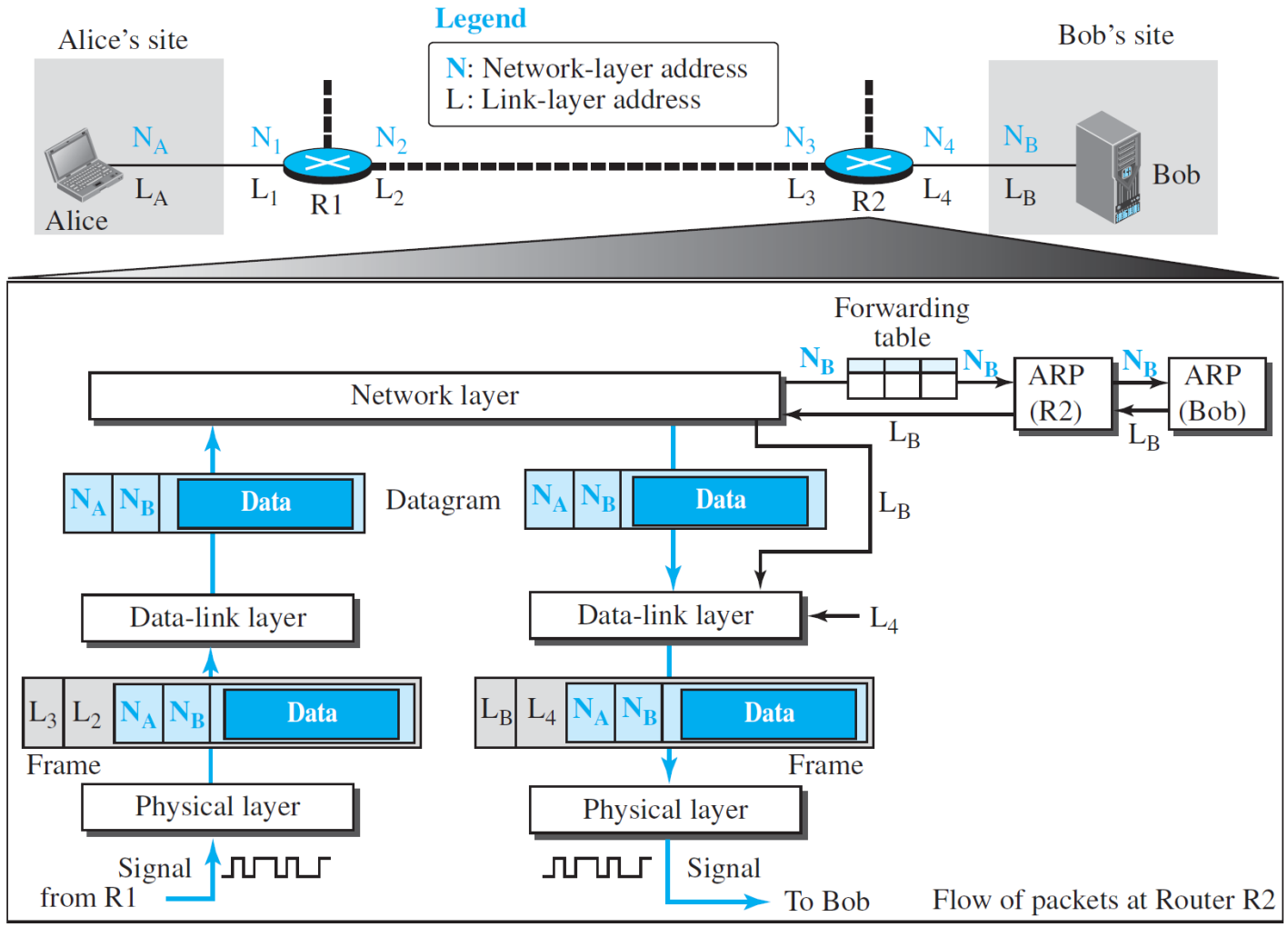
The router R1 receives the frame and try to forward to router R2. R1 needs to know the MAC address of R2 in order to send data in the data link layer.




- The router ($R_1$) receives the frame and try to forward to router ($R_2$).
- $R_1$ needs to know the MAC address ($L_3$) of $R_2$ in order to send data in the data link layer.

# ARP Operations in the Different Networks Internet Communication Example (Step 3)

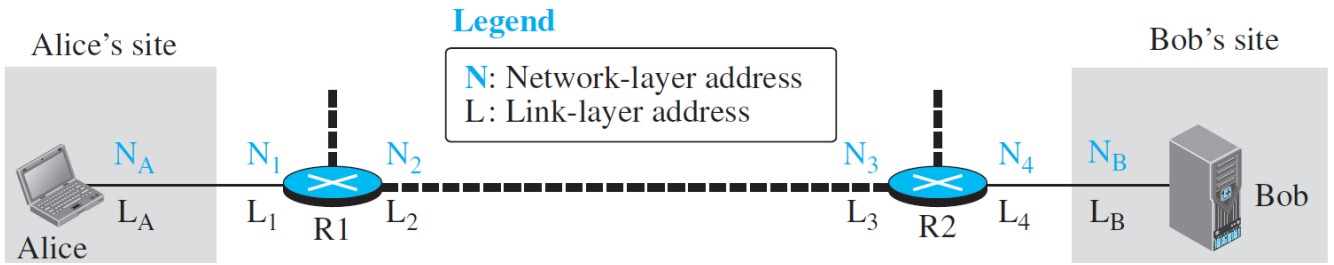
Router R2 receives the frame and have Bob's IP address. The sender and R1 do not know Bob's MAC address because they don't need to know. It is the router R2's job to use ARP to change Bob's IP address to Bob's MAC address.

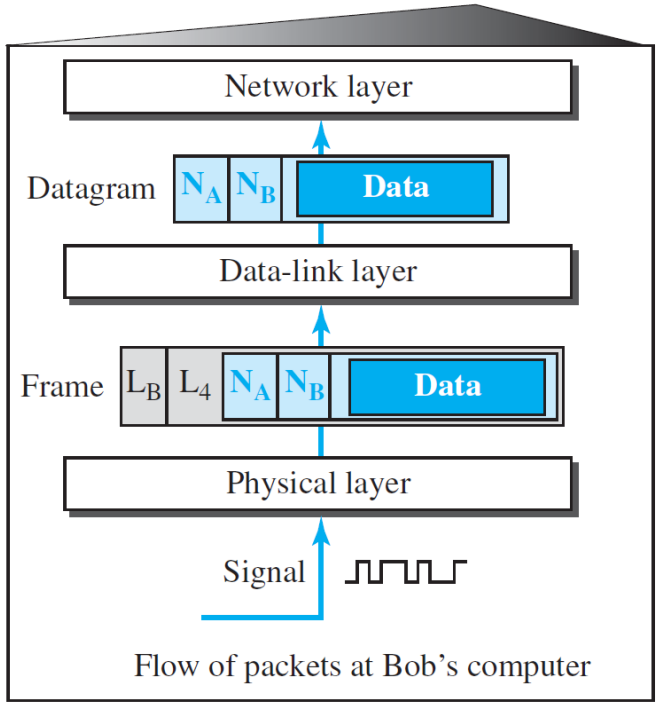



- Router $R_2$ receives the frame and have Bob's IP address ($N_B$). The sender ($R_2$) and $R_1$ do not know Bob's MAC address ($L_B$) because they don't need to know.
- It is the router $R_2$'s job to use ARP to change Bob's IP address ($N_B$) to Bob's MAC address ($L_B$) .

# ARP Operations in the Different Networks Internet Communication Example (Step 4)

Bob checks whether the frame received contains correct information.

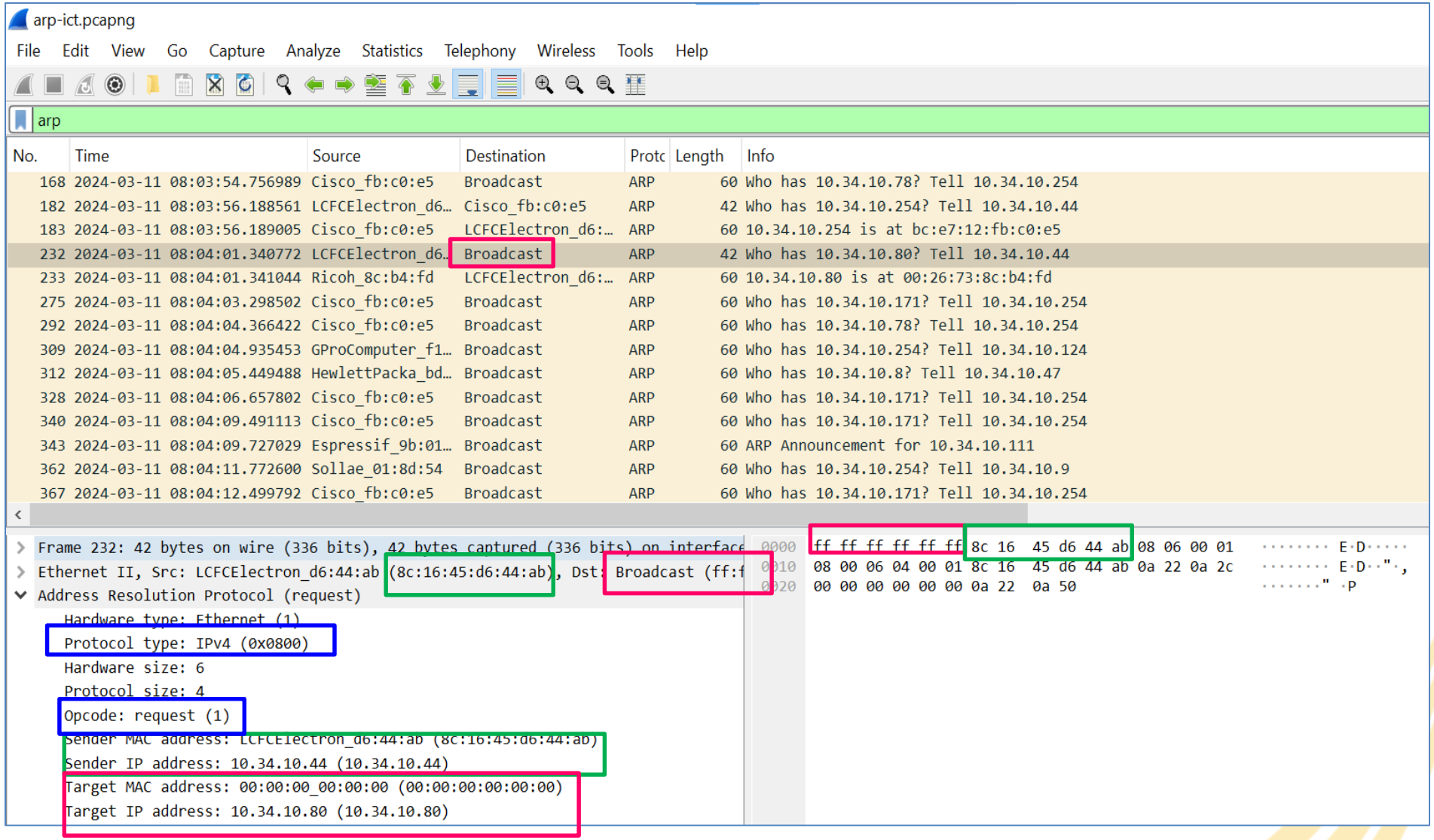



- Bob checks whether the frame received contains correct information.

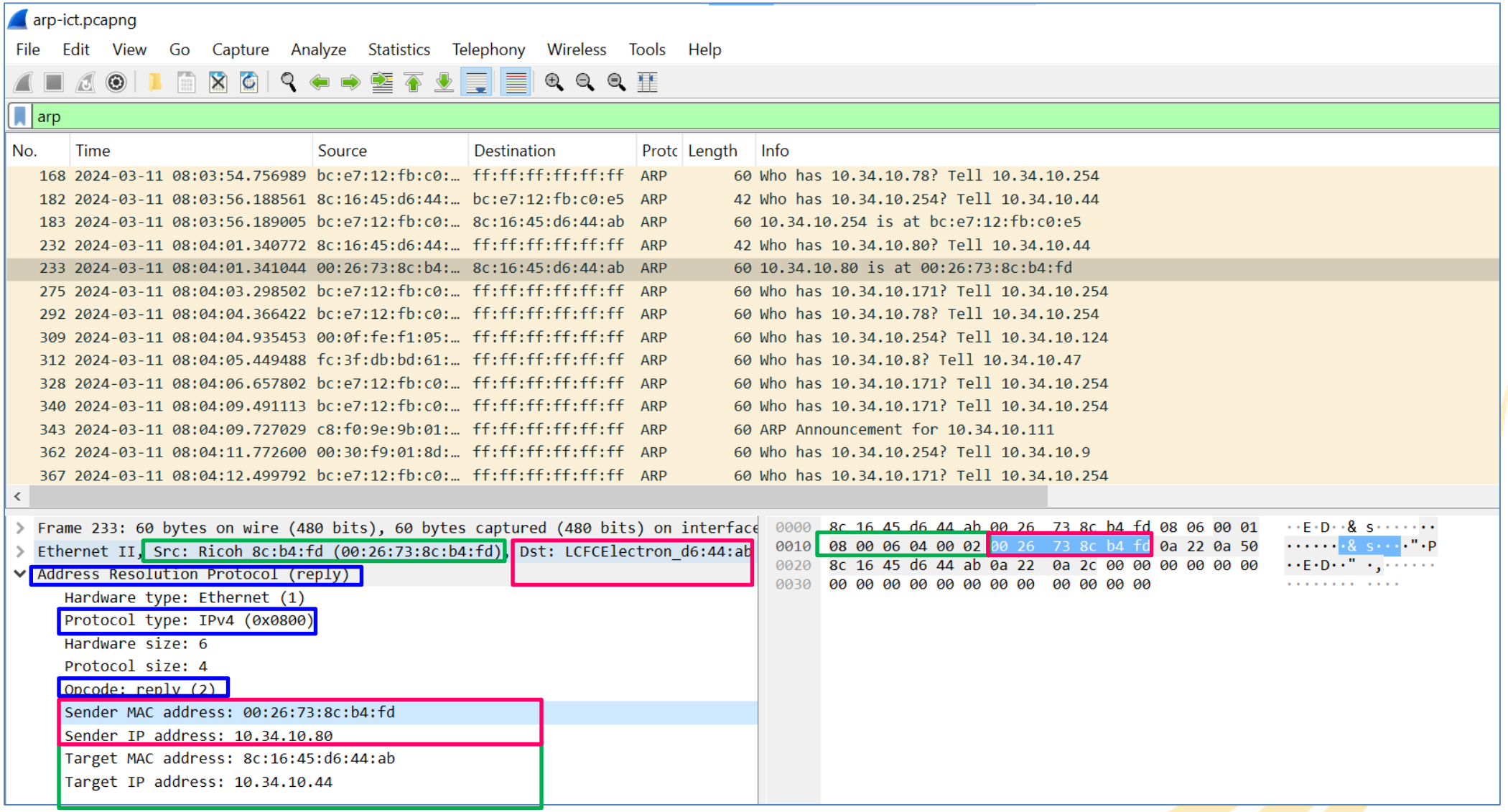
# Wireshark

## ARP Request



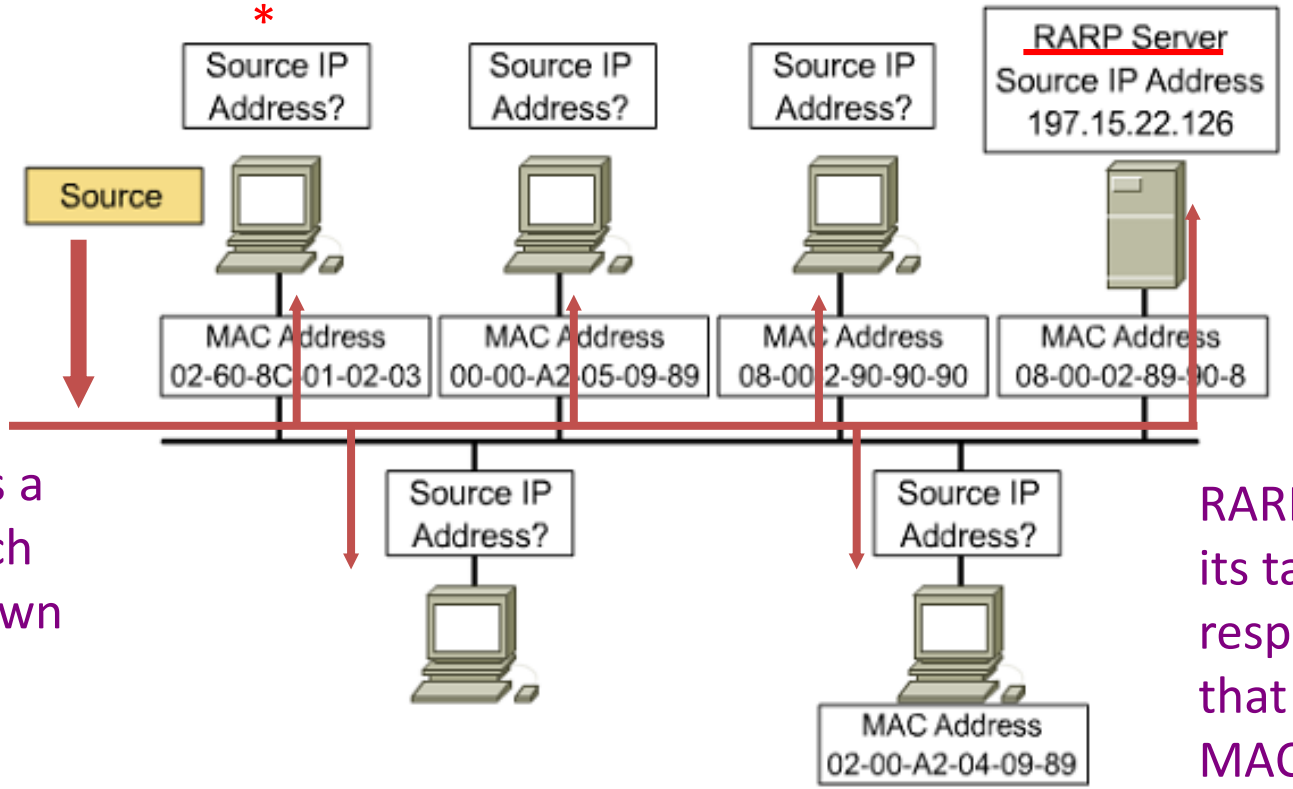
# Wireshark

# ARP Reply



# RARP Request/Reply (Ask for IP Address)

✓https://wiki.wireshark.org/uploads/5786c3aca4f163fdbdcf55d92da6c3f2/rarp_req_reply.pcapng




The source initiates a RARP request, which helps it detect its own IP address.

RARP Server checks its tables and responds with the IP that matches the MAC

| MAC HEADER | IP HEADER | RARP REQUEST MESSAGE |
|---|---|---|
| Destination<br>08-00-02-89-90-8<br>Source<br>02-60-8C-01-02-03 | Destination<br>11111111<br>Source<br>???????? | What is my IP address? |

# Let’s do the Exercises Part 2